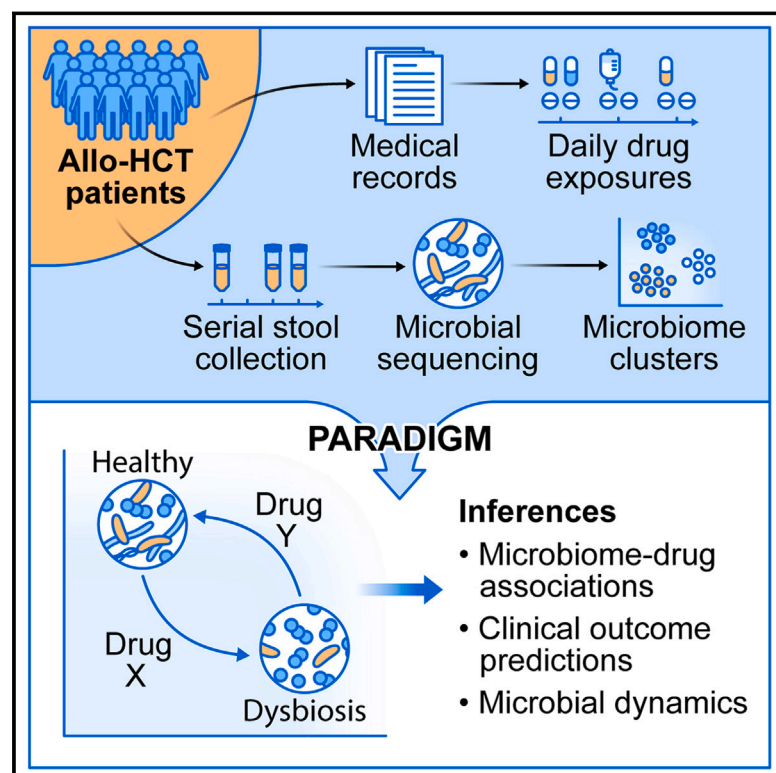


High-resolution analyses of associations between medications, microbiome, and mortality in cancer patients

Graphical abstract



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Highlights

- A computational method identifies medication-microbiome associations
- Applied to the dataset of cancer patients' fecal microbiome profiles and medication records
- *In silico* results recapitulate *in vitro* measurements of anti-bacterial activities
- Medication-bacteria associations are predictive of patient clinical outcomes



Article

High-resolution analyses of associations between medications, microbiome, and mortality in cancer patients

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SUMMARY

Discerning the effect of pharmacological exposures on intestinal bacterial communities in cancer patients is challenging. Here, we deconvoluted the relationship between drug exposures and changes in microbial composition by developing and applying a new computational method, PARADIGM (parameters associated with dynamics of gut microbiota), to a large set of longitudinal fecal microbiome profiles with detailed medication-administration records from patients undergoing allogeneic hematopoietic cell transplantation. We observed that several non-antibiotic drugs, including laxatives, antiemetics, and opioids, are associated with increased *Enterococcus* relative abundance and decreased alpha diversity. Shotgun metagenomic sequencing further demonstrated subspecies competition, leading to increased dominant-strain genetic convergence during allo-HCT that is significantly associated with antibiotic exposures. We integrated drug-microbiome associations to predict clinical outcomes in two validation cohorts on the basis of drug exposures alone, suggesting that this approach can generate biologically and clinically relevant insights into how pharmacological exposures can perturb or preserve microbiota composition.

The application of a computational method called PARADIGM to a large dataset of cancer patients' longitudinal fecal specimens and detailed daily medication records reveals associations between drug exposures and the intestinal microbiota that recapitulate *in vitro* findings and are also predictive of clinical outcomes.

INTRODUCTION

Gut microbiota perturbations have been associated with various diseases and frequently linked to environmental exposures

including antibiotic use and nutritional deficiencies.^{1,2} Non-antibiotic drugs can also contribute to intestinal microbiota changes,³ but their effects in humans are less well-understood and challenging to study due to a lack of reliable drug-exposure data



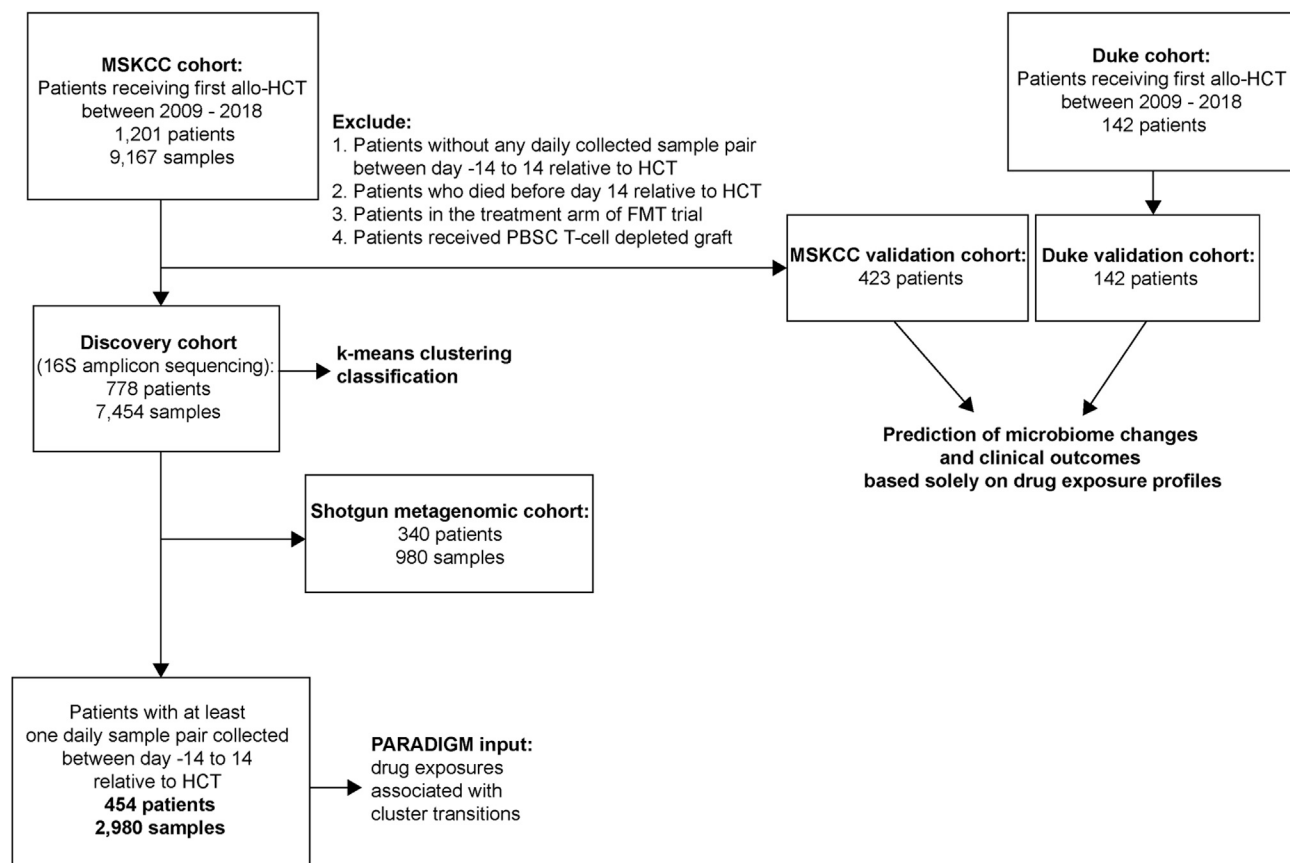


Figure 1. Patient selection criteria for the discovery and validation cohorts

The MSKCC discovery cohort was included in the clustering of sequencing data and PARADIGM algorithm training set. The validation cohorts were included in the analysis of clinical outcomes.

See also [Figure S1](#).

(e.g., recall-based surveys of habitual use of chronic medications)^{1,4} and the absence of densely collected longitudinal fecal samples.^{5,6} Moreover, several pioneering studies of medication exposures and microbiome composition focused on volunteers at relatively healthy steady states.^{4,5} Patients undergoing allogeneic hematopoietic cell transplantation (allo-HCT) exhibit major perturbations of fecal microbiome composition throughout treatment course that have been associated with increased risk of mortality.^{7–9} These patients are exposed to a variety of drugs upon prolonged hospitalizations, during which a wealth of data is routinely gathered as part of their electronic health records. As such, this patient population, from which we have assembled a large bank of fecal specimens, presents a unique opportunity to investigate intestinal microbial responses to drug exposures *in vivo*.

Previous studies in these patients have largely focused on the effect of antibiotics on the intestinal microbiota^{8,10,11}; however, many non-antibiotic drugs routinely administered during this treatment have demonstrated anti-bacterial activities *in vitro*.³ Furthermore, the microbiome perturbations in allo-HCT patients are observed prior to the administration of broad-spectrum antibiotics, suggesting a potential influence of transplant-associated cancer treatment and supportive-care medications.¹² Here, we

inferred relationships between medications, microbiome composition, and clinical outcomes by developing, applying, and validating a new computational method we named PARADIGM (parameters associated with dynamics of gut microbiota) to a large dataset of 16S rRNA and shotgun metagenomic sequencing profiles of serially collected fecal samples from allo-HCT patients.

RESULTS

Clustering captures the temporal dynamics of the intestinal microbiome during allo-HCT

The dataset consists of 9,167 fecal samples from 1,201 allo-HCT patients at Memorial Sloan Kettering Cancer Center (MSKCC; [Figure 1](#); [Table 1](#)). We divided the MSKCC cohort into discovery (7,454 samples; 778 patients) and validation (1,713 samples; 423 patients) cohorts ([Figure 1](#); [Table 1](#)). We computed the compositional differences among discovery cohort samples using Bray-Curtis β -diversity dissimilarity indices at the genus level for 16S rRNA-sequenced samples or at the species level for shotgun metagenomic sequenced samples and visualized the high-dimensional stool composition data via t-stochastic neighbor embedding (tSNE; [Figures 2A and 2B](#)).

Table 1. Patient and sample cohorts from MSKCC and Duke

	MSKCC discovery	MSKCC validation	Duke validation
Number of patients	778	423	142
Number of patients with 16S-sequenced samples	778	405	138
Number of 16S-sequenced samples	7,454	1,713	473
Number of 16S-sequenced samples per patient, median (first-third quartile)	7 (3–14)	3 (2–5)	2 (1–5)
Number of patients with at least one pair of daily 16S-sequenced samples between day –14 and 14 relative to HCT	454	–	–
Number of pairs of daily 16S-sequenced samples between day –14 and 14 relative to HCT	2,039	–	–
Number of pairs of daily 16S-sequenced samples per patient, median (first-third quartile)	3 (1–7)	–	–
Number of patients with shotgun metagenomic samples	340	142	–
Number of shotgun metagenomic samples	980	200	–
Number of patients with drug exposures data	775	423	142

We observed patterns of microbiome injuries, including the loss of alpha diversity and enrichment of potentially pathogenic bacteria such as *Enterococcus* and Enterobacteriaceae (Figures 2A and 2B). As has been well-described in allo-HCT patients, these domination events can be profound, to the point of a single taxon comprising >90% of the relative abundance of a fecal sample,^{8,10} and are predictive of specific deleterious clinical outcomes such as bloodstream infections, graft-versus-host disease (GVHD), and mortality.^{9,10,13} A subset of 980 samples with shotgun metagenomic profiling also showed similar patterns of microbiome injuries during allo-HCT (Figure 2B). Specifically, we observed a cluster of samples whose most abundant organisms were various strict anaerobes such as *Ruminococcus gnavus* or *Erysipelatoclostridium ramosum*, as well as distinct clusters enriched for potentially pathogenic facultative species including *Enterococcus faecium*, *Klebsiella pneumoniae*, and *Escherichia coli*.

The reproducibly observed microbiome perturbations in allo-HCT patients offer a unique opportunity to understand dynamics and evolution of relatively distinct perturbed microbiome compositions or states under environmental exposures, in contrast to the more fluid and non-discrete microbiota in healthy populations.^{14–18} Given the mathematical challenge of reducing dimen-

sionality complexity while preserving bacterial community structure, we performed unsupervised k-means clustering on the Bray-Curtis β -diversity matrix of samples in the discovery set and identified ten distinct microbiota clusters (Figure 2C). We also explored other clustering approaches including hierarchical clustering (Figures S1A–S1C) and Dirichlet Multinomial Mixtures (Figures S1D–S1F).¹⁹ Since k-means clustering partitioned samples more evenly (Figure S1G), we utilized k-means clusters for our subsequent analyses.

Lachnospiraceae and *Clostridiales*, which constitute major commensal taxa present in the healthy human gut, were commonly observed in clusters 1–3, which were also characterized by high alpha diversity (when compared with the median diversity of the discovery cohort) (Figures 2D–2F).²⁰ Clusters 7–10 represented low-diversity “dysbiotic” states (Figures 2D–2F). Intestinal domination by a single bacterial organism ($\geq 30\%$ relative abundance) is a hallmark of severe intestinal dysbiosis.¹⁰ *Lactobacillus*, Proteobacteria, and *Streptococcus* were highly enriched in samples in clusters 5, 8, and 9. Clusters 7 and 10 consisted of *Enterococcus*-dominated samples, with cluster 10 specifically enriched for *E. faecium* (Figures 2D and 2E). These compositional clusters also captured the temporal dynamics of the intestinal microbiota during allo-HCT: high-diversity clusters 1–3 were common pre-allo-HCT, whereas low-diversity states, particularly clusters 7–10, were more prevalent after allo-HCT (Figure 2G).

This classification of samples into discrete microbiome states enabled us to model the complex changes in microbial communities as cluster transition probabilities. We observed the transition frequencies of consecutive samples collected at most 7 days apart between day –16 and day 1,084 relative to HCT. Patients remained in the same cluster over two consecutive samples in 2,987 (54.5%) pairs of samples (Figure 2H; Table S1). Patients were less likely to remain in high-diversity clusters 1–3 among consecutive samples (mean frequency 46.4%; SD 5.0%), compared with dominated clusters such as 8–10 that are highly stable (mean frequency 65.1%; SD 14.1%). We observed a significant and negative association between cluster alpha diversity and self-transition probabilities, suggesting that high-diversity clusters are less stable compared with low-diversity clusters (Figure S1J). Transitions from one cluster to another were observed in 2,495 pairs of samples (45.5%).

We observed a particularly strong stability of the *Enterococcus*-high cluster 10, which is of interest due to the association of this genus with poor clinical outcomes following HCT.^{7,9,10} To investigate potential drivers of *Enterococcus* domination stability, we developed a logistic regression model with lasso penalty analyzing cluster 10 stability as a function of parameters including antibiotic exposure, time of sample collection, alpha diversity, and relative abundance of top 20 most abundant genera in cluster 10 (STAR Methods). We applied this model to a dataset of daily sample pairs collected between days –14 and 100 relative to HCT and found that higher relative abundances of *Staphylococcus* and *Erysipelatoclostridium* were associated with decreased cluster 10 stability (Figure S1K). On the other hand, a higher relative abundance of *Enterococcus* was associated with increased cluster 10 stability, indicating that *Enterococcus* domination leads to a positive feedback

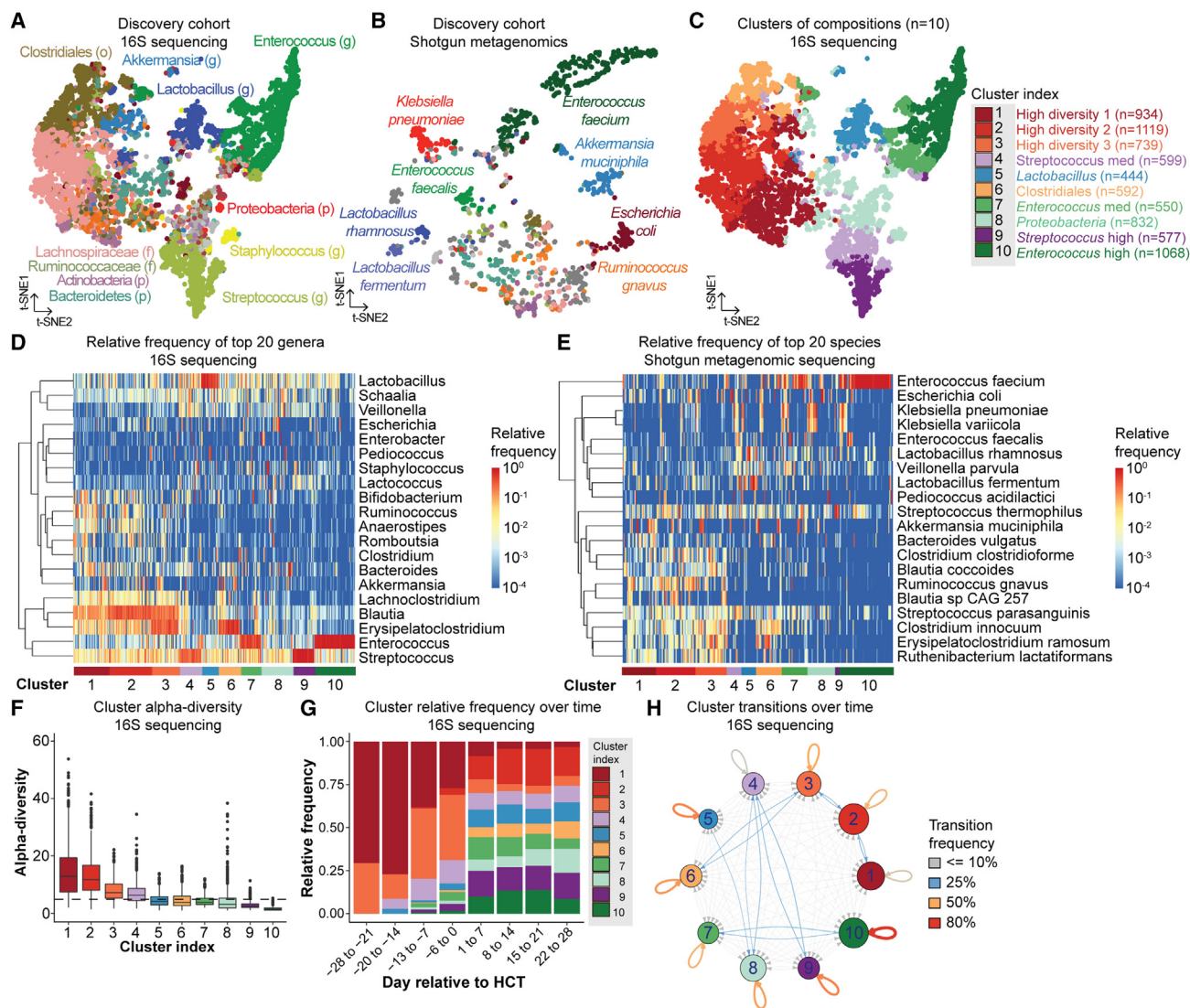


Figure 2. The intestinal microbiota of allo-HCT patients is highly dynamic

(A and B) Compositional space of the intestinal microbiota in the MSKCC discovery cohort visualized by t-SNE projection. Each point represents a sample, colored according to the taxon of highest relative abundance based on (A) 16S rRNA (7,454 samples; 778 patients) or (B) shotgun metagenomic sequencing profiles (980 samples; 340 patients) (p, phylum; f, family; o, order; g, genus). Samples were collected between day –30 and 2,205 relative to HCT.

(C) Ten clusters of intestinal microbiome compositions are assigned by k-means unsupervised clustering.

(D and E) Relative abundance of the top 20 most observed (D) genera in the 16S rRNA profiles and (E) species in the shotgun metagenomic profiles in the MSKCC discovery cohort. Each column is one sample; each row is one genus or species. Rows are clustered by hierarchical clustering.

(F) Cluster alpha diversity (reciprocal Simpson index). The horizontal dashed line represents the median alpha diversity of the MSKCC discovery cohort. Box plot includes a mid-line, upper hinge, and lower hinge, which represent the median and third and first quartiles, respectively.

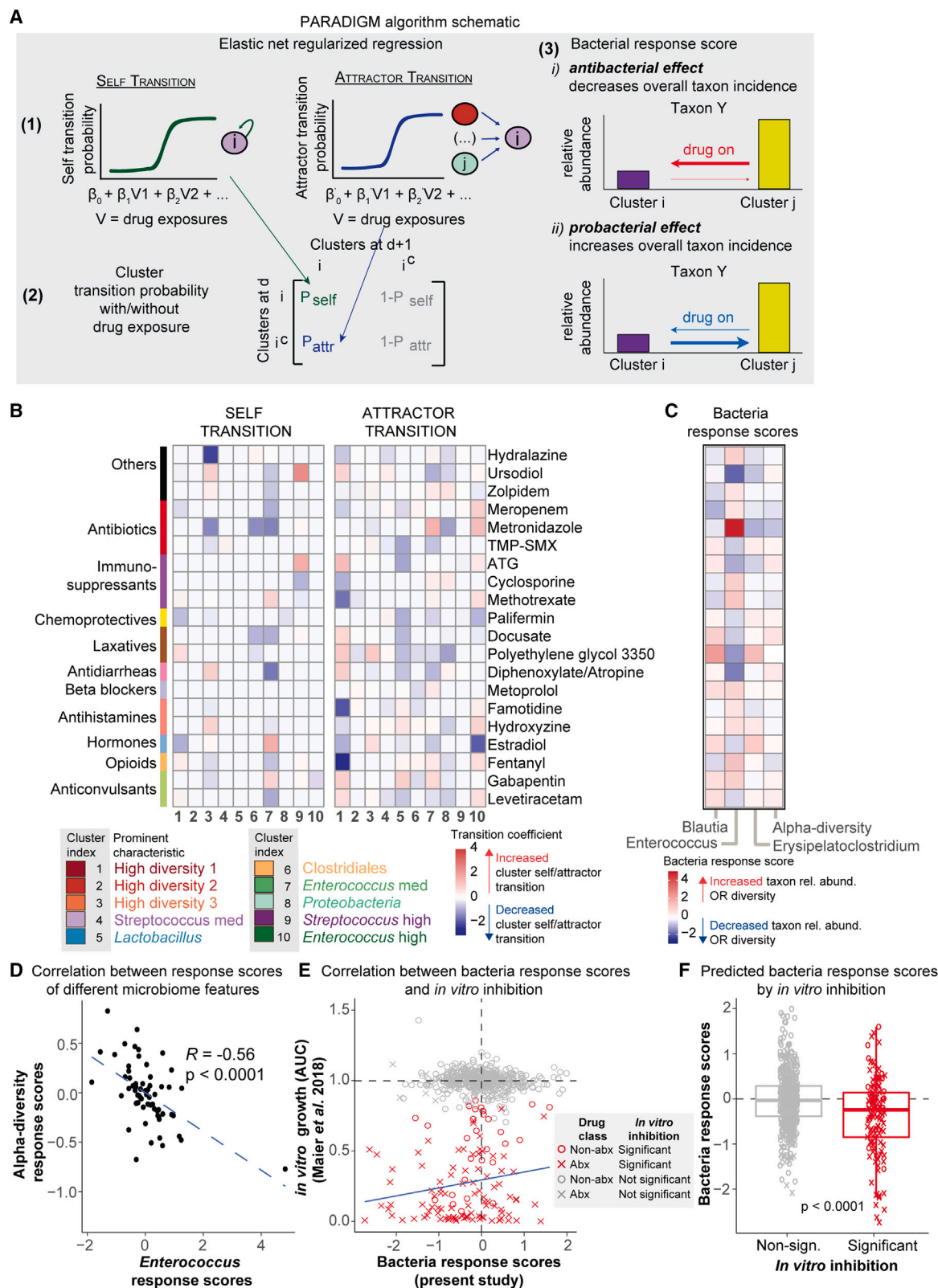
(G) Cluster relative frequency over time relative to HCT.

(H) Network map depicting the transitions among the ten intestinal microbiota clusters over time (5,482 pairs of subsequent samples; 677 patients; collection between day –16 and 1,084 relative to HCT). The thickness of the line is proportional to transition frequency, whereas the node size is proportional to the number of samples per cluster.

loop that supports its own stability. As expected, antibiotic exposure was associated with increased cluster 10 stability. Here, using real-world data, we showed that both environmental factors such as medication exposures, and ecological relationships between bacteria, contribute to microbiota community stability, specifically regarding *Enterococcus* domination.

Non-antibiotic exposures during allo-HCT are associated with changes in the intestinal microbiome compositions

To investigate the associations between drug exposures and microbiome cluster transitions, we developed PARADIGM, a computational tool based on a logistic regression model



(legend on next page)

integrated with first-order Markov-chain transitions (STAR Methods). Markov-chain models have been utilized to investigate microbiome dynamics previously, but the associations between Markov transitions and environmental factors have not been extensively studied.^{14,21–24} The model takes advantage of the high-resolution daily 16S rRNA-sequenced fecal samples (2,039 sample pairs; 454 patients; Table S2) to infer associations between drug exposures and cluster transitions (Figure 3A). For each cluster, we defined two transition types: self-transitions (patients stay in the same cluster) and attractor transitions (patients move to a given cluster). The naming of “self-transition” and “attractor transition” is motivated due to the intuitive connotation they convey about our cluster dynamics. Self-transition describes the probability of a given cluster preserving its current state, and attractor transition describes the probability of a given cluster receiving transitions from any clusters other than itself, in a pair of daily collected samples.

We focused on 62 drugs to which patients in the discovery set were commonly exposed (STAR Methods). To determine the contribution of each drug to the likelihood of self and attractor transitions, we utilized elastic-net logistic regression, where the resulting coefficients for each drug indicate both the direction and magnitude of the association between the drug and daily cluster transitions. Microbiome injury patterns in allo-HCT are strongly linked with time relative to allo-HCT.^{8,10,25} Therefore, we included time as a co-variable to address the temporal patterns of drug exposures and attenuate time-dependent variability in microbiome cluster dynamics (Figure S2). To account for the unequal patient contribution of data points to the training set (Table S2), we pre-specified the 10-fold cross-validation partitions in our training cohort such that samples of the same patient are always in the same partition.

We identified several associations between drug exposures and cluster self and attractor transitions (Figure 3B; full results in Figure S3). As expected, several antibiotics used to empirically treat neutropenic fever were associated with profound changes in the intestinal microbiota. Exposures to meropenem (2.5-fold increase) and metronidazole (3.4-fold increase) were associated with increased transitions to the *Enterococcus*-high cluster 10, consistent with previous reports.^{10,26} Several non-antibiotic medications were also associated with specific cluster dynamics. Exposure to aprepitant, a tachykinin receptor antagonist used for chemotherapy-induced nausea, was associated with a 2.8-fold

increase in transition frequency to the *Enterococcus*-high cluster 10 (Figure S3). Similarly, exposure to the opioid analgesic fentanyl was associated with a 1.9-fold increase in transition frequency to the *Enterococcus*-high cluster 10. Other medications such as labetalol and insulin, which are not known to target intestinal bacteria, were associated with decreased stability of the *Enterococcus*-high cluster 10 (Figure S3). Altogether, PARADIGM identified the association between several non-antibiotic drugs and changes in the intestinal microbiome.

Previous studies have identified specific bacteria that have either beneficial or deleterious associations with clinical outcomes following allo-HCT.^{27–29} As such, we translated drug-cluster associations into drug-taxon associations by calculating bacteria response scores to identify associations between drug exposure and changes in specific taxonomic groups of clinical interests (Figure 3C; full results in Figure S4). In our model, bacteria response scores estimate the association between drug exposure and a microbiome feature, namely the relative abundance of taxa or alpha diversity, where positive scores indicate association with higher relative abundances or diversity values (Figure 3A). We focused on four microbiome features previously associated with allo-HCT patient outcomes, namely the relative abundance of *Enterococcus*, *Blautia*, *Erysipelatoclostridium*, and alpha diversity.^{7–9,28,29} Most antibiotics used in this cohort as empiric or pathogen-directed treatments (metronidazole, meropenem, aztreonam, and cefepime) were associated with an increased relative abundance of *Enterococcus* as well as decreased alpha diversity, consistent with previous studies (Figures 3C and S4).^{10,30,31} Our observation that cefepime exposure was associated with *Enterococcus* expansion is consistent with our previous report³⁰ and may be partly explained by the poor activity of cefepime and other cephalosporins against enterococci. Piperacillin-tazobactam exposures were associated with decreased *Enterococcus* relative abundance, as well as decreased relative abundance of intestinal commensals such as *Blautia* and *Erysipelatoclostridium* to a greater extent compared with other empiric antibiotics (Figure S4). We also observed that drugs most strongly associated with *Enterococcus* expansion were the non-antibiotic drugs including opioids such as fentanyl and hydromorphone, hormones such as levthyroxine, and anticonvulsants such as gabapentin (Figure 3C and S4). Opioid exposures have been previously associated with a decreased relative abundance of *Blautia* in ICU

Figure 3. PARADIGM predicts changes in microbiome features such as genus relative abundance and alpha diversity following drug exposures

- (A) Schematic representation of PARADIGM which takes advantage of daily sampling 16S rRNA-sequenced samples and cluster transitions to infer how drug exposures are associated with microbial dynamics. Bacteria response scores translate drug-cluster associations into drug-genus associations.
- (B) Associations between drug exposures and microbiome cluster dynamics (full results in Figure S3). Self-coefficients indicate whether drug exposure increases or decreases the log-odds of cluster stability. Attractor coefficients indicate whether drug exposure increases or decreases the log-odds of transition to a given cluster. TMP-SMX, sulfamethoxazole/trimethoprim; ATG, anti-thymocyte globulin.
- (C) Bacteria response scores predict the association between a given drug exposure and changes in genus relative abundance or alpha diversity (full results in Figure S4).
- (D) Pearson's correlation between *Enterococcus* response scores and alpha diversity response scores. Each point represents an individual drug.
- (E) Pearson's correlation between bacteria response scores and measurements of *in vitro* inhibition.³ Each point represents the association between a unique drug-species pair.
- (F) Predicted bacteria response scores by *in vitro* inhibition. Two-sided Wilcoxon's rank-sum test. Box plot includes a mid-line, upper hinge, and lower hinge, which represent the median and third and first quartiles, respectively.

See also Figures S2, S3, and S4.

patients who did not receive antibiotics,³² an observation we also reported here for fentanyl and hydromorphone (Figure S3). In contrast, laxatives such as docusate and polyethylene glycol were strongly associated with decreased *Enterococcus* relative abundance (Figure 3C). Previous experimental studies have demonstrated that polyethylene glycol induces global changes in bacterial compositions in mice, either through the modulation of intestinal osmolality or through direct anti-bacterial inhibition.^{33,34} Overall, drug exposures associated with alpha-diversity preservation were correlated with decreased *Enterococcus* expansion and vice versa (Figure 3D).

Validation of *in silico* findings from real-world patient dataset against an independent *in vitro* dataset

We tested the predictive power of PARADIGM by comparing the *in silico* results described here using “real-world data” from cancer patients with an independent published *in vitro* dataset³ (Figures 3E–3F). The *in vitro* screen and the present study share in common 19 bacterial species and 34 drugs. We observed an enrichment of drug-species pairs that showed *in vitro* inhibition (for both antibiotics and non-antibiotics) in the lower left quadrant of the plot, which corresponds to negative response scores *in silico* (Fisher’s Exact Test: odd ratio = 0.60, p value = 0.01). Furthermore, drug-species pairs that showed *in vitro* inhibition had significantly lower response scores in the patient dataset when compared with those that did not show inhibition *in vitro* (Figure 3F). Altogether, these results suggest that PARADIGM can accurately predict *in vitro* anti-bacterial activity of both antibiotics and non-antibiotics and distinguish direct interactions of drugs with bacteria species from the potential confounding influence of the clinical symptoms prompting these drug exposures.

Antibiotic exposure is a strong predictor of subspecies dynamics

Several experimental studies have demonstrated that the bactericidal spectra of several drugs are species- and strain-specific even within the same genus.^{3,35} We therefore explored the associations between drug exposures and changes in relative abundances of species within genera of clinical importance in allo-HCT, using a subset of 980 specimens from 340 patients in the MSKCC discovery cohort with available shotgun metagenomic sequencing profiles (Figure S5). By applying a linear mixed-effects regression model (STAR Methods), we identified associations between drug exposures and changes in bacterial species’ relative abundance. Again, we detected that several drug exposures spanning different drug classes (antibiotics, laxatives, anti-diarrhea, and opioids) were associated with changes in relative abundance of *Blautia coccoides*, *Blautia producta*, *Enterococcus faecalis*, *Enterococcus faecium*, and *Erysipelatoclostridium ramosum*, although these associations were not statistically significant.

Within human intestinal microbiome communities, most species are represented by a single dominant strain,³⁶ although we have previously observed complex strain dynamics during episodes of domination by *E. faecium* in allo-HCT patients.³⁷ For five species within three genera of interest, we identified the sequence signatures of the dominant strains on the basis of marker gene polymorphisms using the StrainPhlAn algo-

rithm³⁶ and calculated tree-based phylogenetic distances between dominant strains in consecutive patient samples (Figure 4A, upper schematic). Small phylogenetic distances between strains in consecutive samples indicate dominant-strain convergence, whereas large distances suggest dominant-strain divergence over time. For most species, and particularly *E. faecium*, the phylogenetic distance between consecutive samples declined over time (Figure 4A, middle row). This temporal pattern suggests reduction of within-species genetic variability across multiple species over time and the rise of a dominant subtype. Furthermore, subtype variability was inversely associated with species relative abundance (Figure 4A, lower row). This correlation might be a consequence of so-called “selective sweeps”^{38,39} by comparatively better-fit strains or loss of variability due to a population bottleneck that may occur during allo-HCT.⁴⁰ Subspecies diversification following a parabolic fitness landscape has been observed in vancomycin-resistant *E. faecium* isolated from longitudinal stool sampling of allo-HCT patients,³⁷ which this method of strain classification focusing on dominant strains might fail to capture. We observed that antibiotic exposure was a significant predictor of dominant-strain genetic convergence within species *E. faecium*, whereas non-antibiotic exposure was not (Figures 4B and 4C).

Drug-microbiome associations are predictive of future microbiome trajectories and clinical outcomes following allo-HCT

Having demonstrated that drug exposures are associated with microbiota changes and in light of previous reports associating fecal microbial composition with allo-HCT clinical outcomes, we next asked whether patterns of drug exposure alone could predict mortality independent of microbiome data. Using drug-exposure data from a separate MSKCC validation cohort, we defined patient-specific response scores as metrics that quantify the net response of microbiome features to drug-exposure profiles (Figure 5A). For example, the patient-specific *Enterococcus* response scores translated patient drug-exposure profiles into relative risk of *Enterococcus* expansion (Figure 5B). We tested these patient-specific bacteria response scores against the outcomes of all-cause and specific-cause mortality in two independent validation cohorts: a subset of 423 MSKCC patients who were not included in the PARADIGM training cohort, as well as 142 patients from an independent cohort from Duke University Medical Center. All 62 drug exposures were considered for the MSKCC validation cohort, whereas only antibiotic exposures were evaluated in the Duke validation cohort. Patient characteristics in these two cohorts are outlined in Table 2.

We observed that patient-specific bacteria response scores based on drug exposures between days –14 to 14 relative to HCT were significantly and positively correlated with observed taxa relative abundance and alpha diversity, respectively, from samples collected between days 14 and 45 relative to HCT in both the MSKCC and Duke validation cohorts (Figure 5C). Furthermore, patients whose drug-exposure profiles predicted higher *Enterococcus* expansion were at an increased risk of all-cause and transplant-related mortality following allo-HCT in the MSKCC and Duke validation cohorts. Specifically, in the

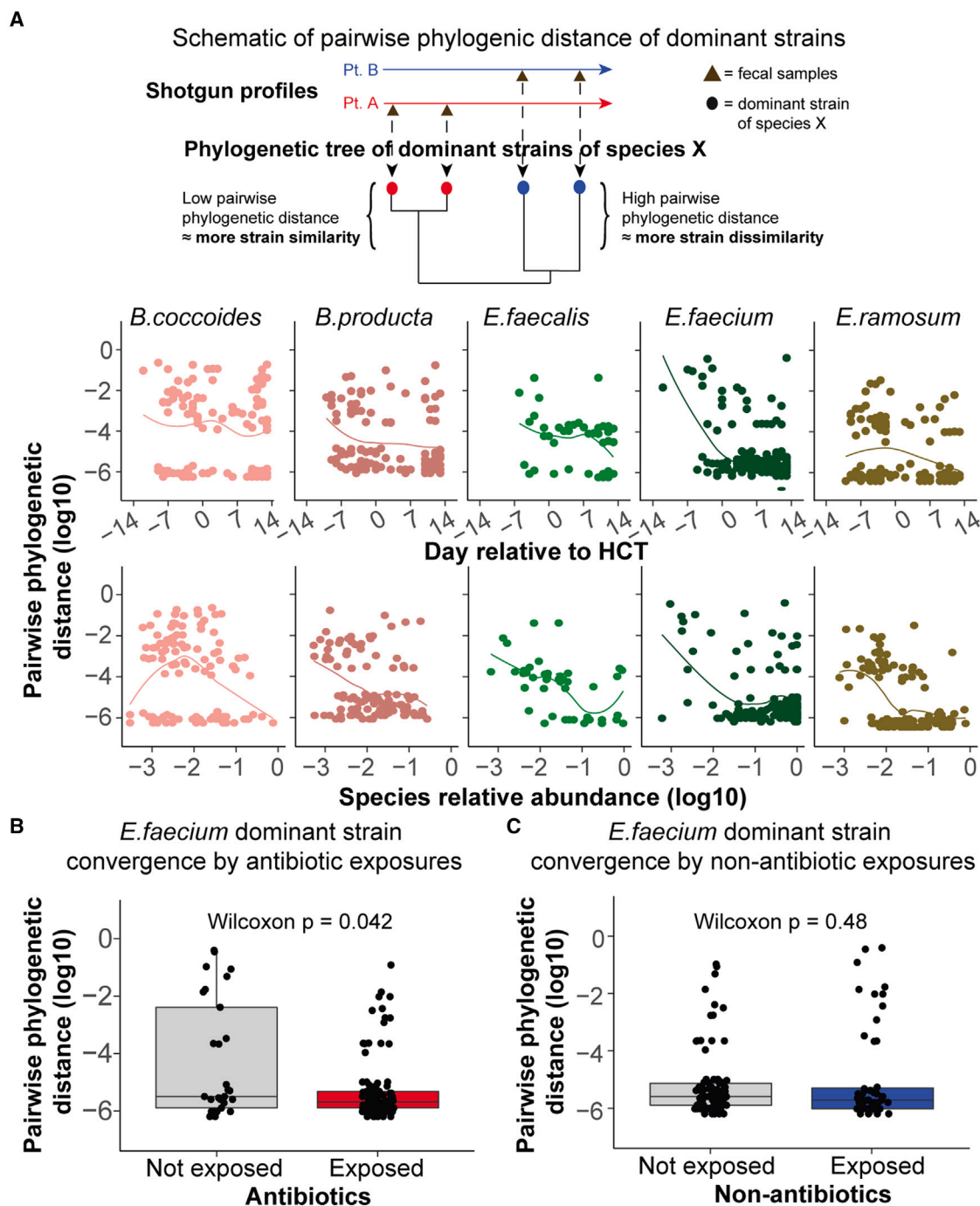


Figure 4. Antibiotics are strong predictors of strain genetic convergence during allo-HCT

(A) Strain convergence over time relative to HCT (middle row), or by species relative abundance (bottom row). Each point represents the tree-based phylogenetic distance between the dominant strains of a given species in a pair of subsequently collected samples. Higher phylogenetic distance suggests genetic dissimilarity, whereas lower phylogenetic distance suggests strain genetic similarity.

(B and C) Antibiotic exposure (B), but not non-antibiotic exposure (C), is associated with increased *E. faecium* dominant-strain convergence. Each point represents the phylogenetic distance between *E. faecium* dominant strains in a pair of subsequently collected samples, stratified by drug exposures during the time gap of sample pair collection. Two-sided Wilcoxon's rank-sum test. Box plot includes a mid-line, upper hinge, and lower hinge, which represent the median and third and first quartiles, respectively.

See also Figure S5.

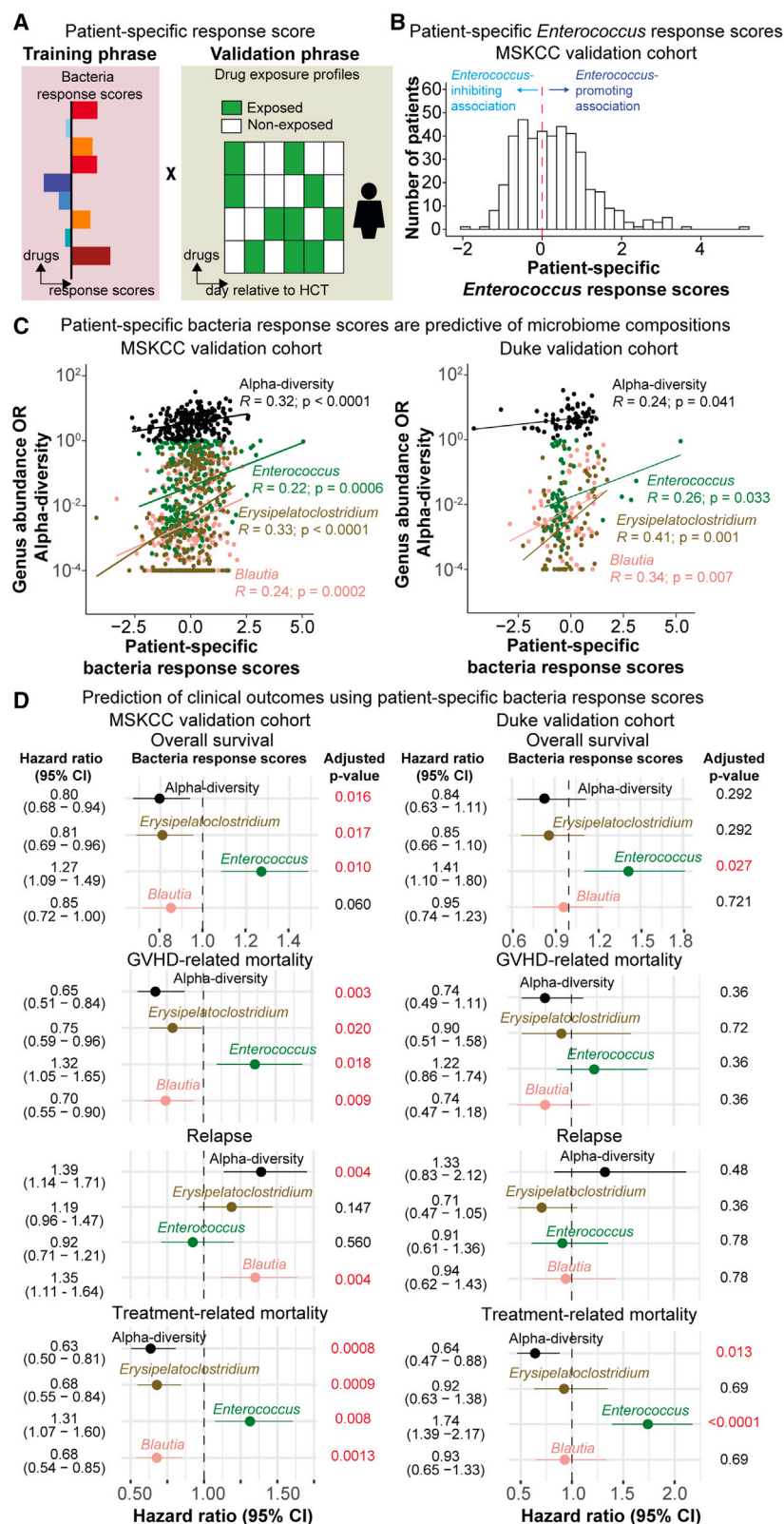


Figure 5. Drug-exposure profiles are predictive of future microbiome trajectories and allo-HCT patient outcomes in two distinct validation cohorts

(A) Schematic of the patient-specific bacteria response score calculation.

(B) Patient-specific *Enterococcus* response scores in the validation cohort were derived based solely on drug-exposure profiles (between day -14 and 14 relative to HCT) and bacteria response scores presented in Figure 3C. A negative score indicates that the drug-exposure profile is associated with an *Enterococcus*-inhibiting effect, whereas a positive score indicates that the drug-exposure profile is associated with an *Enterococcus*-promoting effect.

(C) Pearson's correlation between patient-specific bacteria response scores and observed genus relative abundance or alpha diversity in samples collected between days 14 and 45 relative to HCT in the MSKCC validation cohort (423 patients) and Duke cohort (142 patients). Adjusted p values by Benjamini-Hochberg's correction.

(D) Patient-specific bacteria response scores are predictive of overall and cause-specific mortality in the MSKCC and Duke validation cohorts, in each respective multivariate Cox proportional hazard or Fine-Gray model, controlled for age, sex, conditioning intensity, graft source, and underlying disease. Adjusted p values by Benjamini-Hochberg's correction. The error bar represents the 95% confidence interval.

See also Figure S6.

Table 2. Patient characteristics from MSKCC and Duke cohorts

	MSKCC discovery	MSKCC validation	Duke validation
Number of patients	778	423	142
Mean age at HCT, year (SD)	55 (13)	53 (13)	51 (13)
Year of HCT (%)			
2009–2015	374 (48)	331 (78)	33 (23)
2016–2019	404 (52)	92 (22)	109 (77)
Sex (%)			
Female	312 (40)	166 (39)	43 (30)
Male	466 (60)	257 (61)	99 (70)
Disease (%)			
AML	278 (36)	144 (34)	42 (30)
Others	500 (64)	279 (66)	100 (70)
Conditioning intensity (%)			
Nonmyeloablative	46 (6)	89 (21)	13 (9)
Reduced intensity	171 (22)	245 (58)	6 (4)
Ablative	561 (72)	89 (21)	123 (87)
Median follow-up, months	46	49	11
Graft type (%)			
Cord blood	62 (8)	131 (31)	19 (13)
BM unmodified	78 (10)	30 (7)	13 (9)
PBSC unmodified	176 (23)	262 (62)	110 (78)
PBSC T cell depleted	462 (59)	–	–

IQR, interquartile range; SD, standard deviation; AML, acute myeloid leukemia; BM, bone marrow; PBSC, peripheral blood stem cell; PBSC T cell depletion was performed by *ex vivo* CD34-selection of the graft.

MSKCC validation cohort, patient-specific *Enterococcus* response scores were also significantly associated with an increased risk of GVHD-related mortality (Figure 5D). Conversely, patients whose drug-exposure profiles predicted *Erysipelatoclostridium*, *Blautia*, or alpha-diversity preservation had a decreased risk of all-cause mortality in both the MSKCC discovery and validation cohorts, as well as a decreased risk of GVHD-related mortality in the MSKCC validation cohort (Figures 5D and S6B). Overall, we demonstrated that drug-microbiome associations are predictive of subsequent changes in the intestinal microbiome compositions post-exposure and of clinical outcomes in allo-HCT patients.

The framework for this analysis is a hypothesis that drug exposures affect the intestinal microbiota, which in turn shapes clinical outcomes. In some scenarios, however, patients at high risk for adverse outcomes (for a variety of reasons unrelated to the microbiome) might have received drugs that affect the intestinal bacteria. To explore these possibilities, we focused on *Enterococcus* and compared the hazard ratios of *Enterococcus* relative abundance with the corresponding patient-specific response scores in a Cox proportional hazard ratio model. We reported a significant association between *Enterococcus* abundance and mortality risk, which was stronger than the association between patient-specific *Enterococcus* response scores

and overall mortality, specifically in the MSKCC validation cohort (Figure S6C). However, we also observed that patient-specific response scores remained a statistically significant predictor of mortality risk when controlled for intestinal microbiome compositions (Figure S6C). Altogether, these results suggest that the association between drug exposures and clinical outcomes is partially dependent on drug interactions with the intestinal microbiota.

DISCUSSION

We developed PARADIGM, a computational method that identifies the associations between drug exposures and intestinal microbial dynamics in humans. At its core, PARADIGM analyzes how discrete states of intestinal microbial compositions respond to both antibiotic and non-antibiotic drugs. Although other computational methods have largely focused on antibiotics,^{41,42} our approach further reveals the associations between many non-antibiotic drugs and microbial dynamics in high resolution by analyzing a large dataset of daily stool samples from allo-HCT patients.

Our method was able to infer meaningful associations between drug exposures and microbiome despite the various confounding parameters such as the clinical symptoms prompting these drug exposures, which is particularly important when analyzing medications used to treat gut toxicity. We validated our findings by comparing bacterial response scores derived from this real-world patient study with published *in vitro* observations³ and found that our estimates were significantly correlated with the reported data. Furthermore, the bacterial response scores, calculated based solely on patient drug exposures, allowed us to predict future microbial changes and patient outcomes in two independent validation cohorts, particularly for *Enterococcus* responses. These results demonstrate that PARADIGM can generate hypotheses with both biological and clinical relevance.

In conclusion, we provide insights into the associations between pharmacologic exposures and changes in intestinal microbial composition at the early stage of transplant. The algorithm we have developed, PARADIGM, identifies biologically meaningful and clinically relevant associations between drug exposures and intestinal microbial dynamics in humans. PARADIGM facilitates the integration of drug exposures from many classes of xenobiotic agents, microbiome dynamics, and clinical outcomes to understand the determinants of microbiome health. This computational framework is well-suited for longitudinal data and may be built upon in the future to investigate other environmental parameters of interest, such as dietary intake or other components of the “exposome,” or applied to other disease settings in which drug-microbiome interactions are of clinical importance.⁴³

LIMITATIONS OF THE STUDY

Our study enabled the identification of medication-microbiome associations using a high-throughput approach and showed the power of those associations in predicting clinical outcomes. However, important limitations exist, and the interpretation of the

results should be performed with care. (1) This paper analyzes a retrospective cohort study. Controlled experiments are required to confirm whether our predictions are causal or not. (2) Drug-microbiome interactions are potentially dependent on drug dosing, which we did not account for. (3) Although synergistic and antagonistic drug interactions have been reported,⁴⁴ our model assumes drugs to act independently from each other. (4) Transplant-specific effects, such as conditioning intensity and graft type, are not fully captured by our model. We assume those effects are partially attenuated by a time parameter and the medication protocol depending on the transplant type and conditioning regimen. (5) Medication exposure is only one component among the various perturbations a patient is subjected to. Other environmental factors have also been associated with changes in the intestinal microbiome. For example, dietary intake has been shown to play a major role in shaping the intestinal microbiota in both healthy individuals and cancer patients.^{45–47} Future efforts to collect dietary intake data could help elucidate further the association between diet, changes in the intestinal microbiota, and clinical outcomes of allo-HCT patients. Finally, we envision PARADIGM to be a valuable tool to investigate microbiome dynamics *in vivo* and await its application to other comparable datasets, encompassing drug exposures and other environmental factors, to replicate and extend the results in this study.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.cell.2023.05.007>.

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AUTHOR CONTRIBUTIONS

C.L.N. and A.L.C.G. conceived of and designed the project and wrote the manuscript. C.L.N. designed and performed all computational analyses and analyzed and interpreted the results. K.A.M. and J.U.P. guided analysis and result interpretation and wrote the manuscript. O.M., R.S., S.M.D., and B.G. assisted with result interpretation and provided essential tools and insights. A.D., N.W., K.S., T.F., B.P.T., C.L., and J.B.X. provided computational expertise guidance and assisted with sequencing analysis pipelines. J.B.S., A.E.S., P.A.G., D.G.B., A.G.C., M.A.M., and L.B. coordinated clinical data and sample collection and management. G.K.A., E.F., and A.G. contributed to experimental work, including sample processing, DNA extraction, and sequencing. A.D.S., N.J.C., Y.T., S.A.G., M.A.P., and E.G.P. oversaw the long-term bio-banking effort and provided critical expertise. J.U.P., A.L.C.G., and M.R.M.v.d.B. co-supervised the study.

DECLARATION OF INTERESTS

MSKCC has financial interests relative to Seres Therapeutics. M.R.M.v.d.B. has received research support from Seres Therapeutics; has consulted, received honorarium from, or participated in advisory boards for Seres Therapeutics, WindMIL therapeutics, Rheos, Frazier Healthcare Partners, Nektar Therapeutics, Notch Therapeutics, Forty Seven, Priothera, Ceramedix, Lygenesis, Pluto Immunotherapeutics, Magenta Therapeutics, Merck & Co, and DKMS Medical Council (Board); has IP licensing with Seres Therapeutics and Juno Therapeutics; and has stock options from Seres and Notch Therapeutics. J.U.P. reports research funding, intellectual property fees, and travel reimbursement from Seres Therapeutics and consulting fees from DaVolterra, CSL Behring, and MaaT Pharma; serves on an advisory board of and holds equity in Postbiotics Plus Research; and has filed intellectual property applications related to microbiome. K.A.M. is on the advisory board for and holds stock in PostBiotics Plus and has served in an advisory role and received

honoraria from Incyte. R.S. has served on an advisory board for Medexus. B.G. received research funding from Actinium Pharmaceuticals. E.G.P. serves on the advisory board of Diversigen and has received speaker honoraria from Bristol-Myers Squibb, Celgene, Seres Therapeutics, MedImmune, Novartis, and Ferring Pharmaceuticals; is an inventor on patents related to microbiome; and holds patents that receive royalties from Seres Therapeutics. A.D.S. has received research funding from Merck, Novartis, and Seres; has received honoraria from Abbott Nutrition; has consulted for AVROBIO and Targazyme; and has received research supplies from Clasado and DSM/iHealth. M.A.P. reports honoraria from Adicet, AlloVir, Caribou Biosciences, Celgene, Bristol-Myers Squibb, Equilium, ExeVir, Incyte, Karyopharm, Kite/Gilead, Merck, Miltenyi Biotec, MorphoSys, Nektar Therapeutics, Novartis, Omeros, OrcaBio, Syncoption, VectivBio AG, and Vor Biopharma; serves on DSMBs for Cidara Therapeutics, MediGene, and Sellas Life Sciences and the scientific advisory board of NexImmune; has ownership interests in NexImmune, Omeros, and OrcaBio; and has received institutional research support for clinical trials from Incyte, Kite/Gilead, Miltenyi Biotec, Nektar Therapeutics, and Novartis. N.J.C. is on DSMBs for Fate Therapeutics, Takeda, and Celularity. A.L.C.G. is currently employed by and has stock options at Xbiome Inc.

INCLUSION AND DIVERSITY

One or more of the authors of this paper self-identifies as an underrepresented ethnic minority in their field of research or within their geographical location. One or more of the authors of this paper self-identifies as a member of the LGBTQIA+ community.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Biological samples		
Human Fecal Samples	This study	N/A
Chemicals, peptides, and recombinant proteins		
Phenol:Chloroform:Isoamyl Alcohol (25:24:1)	Fisher Scientific	Cat#: AC327111000
TRIS-EDTA 1X, pH 8.0, biotech grade	Fisher Scientific	Cat#: BP2473500
20% SDS Solution, molecular grade	Fisher Scientific	Cat#: BP1311-1
3M Sodium Acetate, pH 5.2, sterile, Teknova	Fisher Scientific	Cat#: NC0496686
Biospec 0.1mm zirconia-silica beads	Fisher Scientific	Cat#: NC0362415
TE Buffer with 0.1mM EDTA, pH 7.6, sterile	Fisher Scientific	Cat#: NC9491715
Ethanol, 200 proof, 0.125gal	Fisher Scientific	Cat#: 04-355-222
RNase A, 100mg; from bovine pancreas, Roche	Fisher Scientific	Cat#: 50-100-3354
Water, Ultrapure, distilled	Fisher Scientific	Cat#: 10-977-023
Sodium Chloride 1M, Teknova	Fisher Scientific	Cat#: NC1326185
Critical commercial assays		
QIAamp DNA mini kit	Qiagen	Cat#: 51306
Qiaquick PCR Purification Kit	Qiagen	Cat#: 28106
MinElute PCR Purification Kit	Qiagen	Cat#: 28006
Collection tubes, 2 mL	Qiagen	Cat#: 19201
Platinum Taq DNA polymerase, w/buffers	Fisher Scientific	Cat#: 10-966-083
dNTP mix 10mM	Fisher Scientific	Cat#: 18-427-088
Qubit dsDNA broad-range assay kit	Fisher Scientific	Cat#: Q32853
Tapestation D1000 ScreenTape, 112rx	Agilent	Cat#: 5067-5582
Tapestation D1000 Reagents, 112rx	Agilent	Cat#: 5067-5583
Tapestation 96-well plate foil seal	Agilent	Cat#: 5067-5154
Tapestation 96-well sample plate	Agilent	Cat#: 5067-5150
Tapestation Loading Tips	Agilent	Cat#: 5067-5153
Tapestation Optical tube strip caps	Agilent	Cat#: 401425
Tapestation Optical tube strips	Agilent	Cat#: 401428
Tapestation High-Sensitivity D1000 Screen Tape, 112rx	Agilent	Cat#: 5067-5584
Tapestation High-Sensitivity D1000 Reagents, 112rx	Agilent	Cat#: 5067-5585
Agilent Technologies 4200 Tapestation	Agilent	Cat#: G2991BA
KAPA Hyper Prep	Roche Diagnostics	Cat#: 07962363001
AMPure Beads XP	Beckman Coulter	Cat#: A63881
MiSeq Reagent Kit v2 (500-cycles)	Illumina	Cat#: MS-102-2003
Deposited data		
Stool metagenomics (16S rRNA and shotgun) sequencing	This study	https://doi.org/10.6084/m9.figshare.21657806.v1
Oligonucleotides		
16S rRNA gene sequencing forward primer: (563F): 5'-nnnnnnnn-NNNNNNNNNNN-AYTGGGYDTAAAGNG-3'	Integrated DNA Technologies	https://earthmicrobiome.org

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
16S rRNA gene sequencing reverse primer: (926Rb): 5'-nnnnnnnn-NNNNNNNNNN-CCGTCAATTYHTTTRAGT-3	Integrated DNA Technologies	https://earthmicrobiome.org
Software and algorithms		
PARADIGM	This study	https://doi.org/10.5281/zenodo.7818943
R (version 4.1.0)	R Foundation	https://www.r-project.org
DADA2 (version 1.16.0)	Callahan et al. ⁴⁸	https://benjjneb.github.io/dada2/index.html
BBMap (version 38.97)	N/A	https://www.sourceforge.net/projects/bbmap/
KneadData (version 0.7.5)	The Huttenhower Lab	https://huttenhower.sph.harvard.edu/kneaddata
HUMAnN 3.0 (version 3.0.1)	Beghini et al. ⁴⁹	https://huttenhower.sph.harvard.edu/humann/
MetaPhlan 3.0 (version 3.1.0)	Beghini et al. ⁴⁹	https://huttenhower.sph.harvard.edu/metaphlan/
StrainPhlan 3.0	Truong et al. ³⁶	http://segatalab.cibio.unitn.it/tools/strainphlan/

RESOURCE AVAILABILITY**Lead contact**

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Marcel van den Brink (vandenbm@mskcc.org).

Materials availability

This study did not generate new unique reagents.

Data and code availability

16S rRNA and shotgun metagenomic sequencing data have been deposited at the NCBI database and are publicly available as of the date of publication. De-identified patient data necessary to reproduce the results in this manuscript and NCBI SRA accession numbers have been deposited on Figshare and are publicly available as of the date of publication. The DOI is listed in the [key resources table](#). All original code has been deposited at Github and is publicly available as of the date of publication. The DOI is listed in the [key resources table](#). Scripts and data are provided to facilitate reproducibility of all the analyses presented in this manuscript, except for the analysis of clinical outcomes. Patient-level clinical outcomes for MSKCC and Duke cohorts are available via data sharing agreement per institutional policies. Any additional information required to reanalyze the data reported in this paper is available from the [lead contact](#) upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS**Study population of human subjects**

The patient and fecal sample cohort in this study has been described in previous studies.^{8,9,50} Stool samples were collected at two different transplant centers, Memorial Sloan Kettering Cancer Center (MSKCC) from April 2009 to September 2019, and Duke University Medical Center from July 2012 to April 2018. Participants in the observational cohorts at both MSKCC and Duke provided written informed consent for the use of their stool samples and clinical data. The use and analysis of these specimens for this study was approved by Institutional Research Boards at both institutions (MSKCC: #16-834; Duke: PRO0006268 and Pro00050975). Stool samples collected after day -30 relative to a first allo-HCT, and before day -10 relative to a second allogeneic transplant (if applicable) were included in the analysis. A subset of patients from MSKCC participated in a randomized clinical trial of fecal microbiota transplantation (FMT, NCT02269150).²⁵ Stool samples from patients in the control arm, as well as from patients in the FMT arm collected pre-FMT were included for analysis. Post-FMT stool samples were excluded. Detailed patient and fecal sample characteristics for both MSKCC and Duke cohorts are outlined in [Tables 1 and 2](#).

METHOD DETAILS

Stool collection and storage

As DNA extraction procedures, sample-handling environment, sequencing and bioinformatics pipelines are important sources of variability in microbiome data, we minimized bias and institutional batch effects by collecting and freezing samples at each center following the same protocol. All stool samples were collected, aliquoted and frozen at their respective clinical centers; extraction, sequencing, and analyses were performed centrally at MSKCC.

Fecal samples were collected in both inpatient and outpatient settings. At MSKCC, inpatient samples were collected by nursing staff from toilet inverted “hats” into ~100 ml-sized containers at the bedside, promptly delivered to the laboratory via pneumatic tube, and refrigerated at 4°C until aliquoted for long term storage at -80°C. Outpatient stools were collected in the patients’ homes using a commode specimen collection system, after which the entire collection bin was capped, placed inside a biohazard zip lock bag to prevent leakage, and deposited inside a 8 x 6 x 4.25” foam container along with pre-chilled freezer packs. Samples were either brought by the patient to a clinic appointment or shipped directly from the patient’s home to the laboratory via courier.

Upon receipt by the Molecular Microbiology Facility Laboratory at MSKCC, each sample was given a unique ID and recorded into the fecal biobank database. Approximately 0.5 ml of whole stool was aliquoted without any preservative solution into four 2 ml cryovials using a clean disposable spatula and transferred into a -80°C freezer until thawed for DNA processing. Duke samples were aliquoted similarly at Duke, batch-shipped frozen to MSKCC, and extracted and sequenced in the same way as the MSK samples at the Molecular Microbiology Facility Laboratory.

DNA extraction

Bacterial DNA was extracted using an optimized phenol-chloroform protocol to recover nucleic acids from tough microbes commonly present in stool samples, as previously described.^{8,51} Briefly, 200–300 mg of solid or 200–300 µl of liquid stool were aliquoted and the respective wet weight was recorded. Fecal samples were resuspended in 500 µl of extraction buffer (i.e., 0.2 M NaCl, 0.2 M Tris-HCl, pH 8.0, and 20 mM ethylenediaminetetraacetic acid, prepared fresh). The mixture was combined with 0.1 mm zirconia/silica beads (approximately 500 µl), 200 µl of sodium dodecyl sulfate, and 500 µl of phenol-chloroform-isoamyl alcohol (25:24:1 solution). The bacterial cells were lysed by mechanical disruption using a bead-beater for 2 minutes at over than 3000 rpm. Detritus was spun down at 16,000g at 4°C for 5 minutes. The upper aqueous layer was transferred to a clean 1.7 ml tube and mixed with 100 µl of extraction buffer as described above, and 500 µl of phenol-chloroform-isoamyl alcohol (25:24:1). The solutions were homogenized by inversion and spun down again at 16,000g for 5 minutes at 4°C. The upper aqueous layer was recovered, and this process was repeated for a total of three times. After the final round of phenol-chloroform-isoamyl alcohol was completed, 400 µl of the top aqueous layer was combined with 40 µl of sodium acetate and 880 µl of cold 100% ethanol. The mixture was vortexed and then frozen for at least 20 minutes, preferably overnight. After the ethanol freeze incubation period, samples were spun down at 16,000g for 20 minutes at 4°C to pellet the DNA. The upper aqueous layer was aspirated, and the visible DNA pellet was resuspended in 200 µl of TE buffer containing 100 mg/mL of RNase solution, followed by an incubation at 50°C for 20 minutes. The genomic DNA was further purified using the QIAamp DNA mini kit (Qiagen) following the manufacturer instructions. The purified DNA was eluted in 100 µl of ultrapure water and stored at -80°C prior to quantification of DNA yield and subsequent PCR amplification.

Sequencing

16S rRNA V4-V5 barcoded amplification and multiplexing

The amplification, multiplexing and sequencing of 16S rRNA from extracted DNA has been previously reported.²⁵ Briefly, genomic purified DNA was diluted if necessary and 50 ng was used as template during PCR amplification. The V4-V5 region of the 16S rRNA gene was amplified with the primers 563F (5'-nnnnnnnn-NNNNNNNNNN-AYTGGGYDTAAAGNG-3') and 926Rb (5'-nnnnnnnn-NNNNNNNNNN-CCGTCATTYHTTTAGT-3), where ‘N’s represent unique 12-base pair Golay barcodes and ‘n’s represent additional nucleotides to offset the sequencing of the primers.⁵² Duplicate PCR reactions were performed for each sample with 2.5 U of Platinum Taq DNA polymerase and 0.5 mM of forward and reverse primers at 94°C for 3 minutes, followed by 27 cycles of 94°C for 50 seconds, 51°C for 30 seconds and 72°C for 1 minute and a final elongation step at 72°C for 5 minutes. Amplicons were purified using the Qiaquick PCR Purification Kit (Qiagen) after pooling the sample replicates. The purified PCR products were quantified using Agilent Technologies 4200 TapeStation and multiplexed at equimolar amounts. The obtained pools of bar-coded 16S amplicons went to further processing for library preparation and sequencing on an Illumina MiSeq platform at paired-end 250 base pair (bp) at the MSKCC Integrated Genomics Operation sequencing core. Extraction blanks were included in each extraction batch as negative controls. These blanks were PCR-amplified but did not show PCR products and were subsequently removed from demultiplexing and sequencing.

Shotgun metagenomic sequencing

Shotgun metagenomic sequencing was conducted as previously described.³⁷ Extracted DNA was sheared to a target size of 650 bps using a Covaris ultrasonicator. DNA was then prepared for sequencing using the Illumina TruSeq DNA library preparation kit and sequenced using the Illumina platform targeting approximately 10–20 million reads per sample with 100-bp paired-end reads.

QUANTIFICATION AND STATISTICAL ANALYSIS

Sequencing bioinformatics pipeline

16S rRNA gene sequencing

Reads were quality-filtered, deduplicated, denoised and amplicon sequence variants (ASVs) were inferred following the DADA2 pipeline (Divisive Amplicon Denoising Algorithm).⁴⁸ Reads were truncated at a length of 180 bp for forward and reverse reads to ensure sufficient quality and length to use overlapping paired end reads. Default values were used for filtering and trimming reads prior to inferring sequence variants through function *filterAndTrim()* (maxN=0, maxEE=2, trunQ=2). Reads per sample were capped at 100,000 prior to sequence variance inference. Samples with more than 100,000 reads were sub-sampled. Each sequencing run was analyzed separately before merging the ASV counts across multiple runs.

Samples with poor sequencing quality were filtered out using the following criteria:

1. Less than 30% reads remaining after filtering and trimming
2. Less than 1,000 reads remaining after filtering and trimming
3. More than 5% adapter contamination

Quality control details are summarized in Table S3. Taxonomic classification was annotated according to the NCBI 16S rRNA sequence database. Alpha diversity was calculated using the reciprocal Simpson index at the ASV level, which is the metric used for all alpha diversity measures in this study. Beta diversity was computed according to the Bray-Curtis distances at the genus level using the *beta_diversity.py* script in the QIIME bioinformatic pipeline.⁵³ We performed t-distributed stochastic neighbor embedding (tSNE) dimensionality reduction for visualization of the intestinal microbiota compositions using a Bray-Curtis β -diversity matrix at the genus level with R package Rtsne (max_iter = 10,000; perplexity = 75; theta = 0.2).

Shotgun metagenomic sequencing

The right and left side of a read in a pair was trimmed to Q10 using the Phred algorithm, using the *bbduk.sh* script in the BBDMap package (<https://www.sourceforge.net/projects/bbmap/>). A pair of reads was dropped if either read had a length shorter than 51 nucleotides after trimming. The 3'-end adapters were trimmed using a kmer of length 31, and a shorter kmer of 9 at the other end of the read. One mismatch was allowed in this process, and adapter trimming was based on pair overlap detection (which does not require known adapter sequences) using the 'tbo' parameter. The 'tpe' parameter was used to trim the pair of reads to the same length. Removal of human contamination was done using Kneaddata with paired end reads, employing BMTagger. The BMTagger database was built with human genome assembly GRCh38.

Samples with poor sequencing quality were filtered out using the following criteria:

1. Less than 50% reads remaining after trimming with BBDMap
2. Less than 1 million reads remaining after trimming
3. More than 5% adapter contamination

After decontamination, the paired-end reads were concatenated to a single FASTQ file as the input for functional profiling with the HUMAnN 3.0 pipeline.⁴⁹ After aligning to the updated ChocoPhlAn and UniRef90 database with default settings, the samples were renormalized by library depth to copies per million. MetaPhlAn 3.0 was used to identify taxonomic compositions with the relative abundance of the species using parameter -t rel_ab.⁴⁹ Similar to the 16S rRNA-sequenced data, we performed tSNE analysis for visualization of shotgun metagenomic compositions using a Bray-Curtis β -diversity matrix at the species level with R package Rtsne (max_iter = 3,000; perplexity = 20; theta = 0.1).

Clustering of the intestinal microbiota compositions

Three unsupervised clustering methods were used to identify distinct clusters of the intestinal microbiota compositions within the MSKCC validation cohort. Unsupervised k-means clustering and hierarchical clustering methods were applied to the Bray-Curtis β -diversity matrix. Dirichlet Multinomial Mixture (DMM) was applied to the raw count matrix at the genus level. For all three methods, cluster size parameter from 2 to 15 clusters was evaluated, ultimately choosing ten clusters for downstream analysis. Since k-means clustering partitioned samples more evenly (Figure S1G), we utilized k-means clusters for our subsequent analyses. We determined ten clusters to be a good balance between reducing the complexity of the 16S rRNA sequencing data while still representing the variability of the intestinal microbiome compositions during allo-HCT.

Domination threshold

We defined sample domination by a single taxonomic units using the threshold of ≥ 0.3 of the 16S rRNA sequencing relative abundance.⁸ The taxonomic color scheme was adapted from the R package yingtool2 (<https://github.com/ying14/yingtools2>) and a previous publication.⁸

Derivation of a mathematical model of microbial dynamics termed PARADIGM

A biologically motivated, simplified mathematical model termed PARADIGM was developed to model microbial dynamics while simultaneously considering the effects of drug exposures on the intestinal microbiota. A naïve model with n possible clusters in a time interval Δt for L distinct drugs would require $\Delta t \cdot n^2 2^L$ parameters, which would quickly saturate the amount of data available in most practical applications. The following characteristics of intestinal microbial dynamics during allo-HCT were considered to simplify our model: (i) The microbial composition is more likely to stay in same cluster within a day window; (ii) Drug exposures perturb clusters differently; (iii) Transitions are easier to occur among close clusters compared to distant clusters; and (iv) Transition events vary over time. As such, we developed a model that include parameters for time, drug exposures and distance between clusters.

We defined microbiota dynamics in terms of two possibilities: the rate at which the microbiota composition stays in the same cluster (self transition) and the rate at which it attracts transitions from other clusters (attractor transition). Formally, attractor transition is a force, measured in terms of probability, that describes the magnitude in which a given cluster may receive transitions from any clusters other than itself, in a pair of daily collected samples. For example, the attractor transition to Cluster 1 is defined as the patient moves to cluster 1 at time t_{d+1} , from any clusters between 2 to 10 at time t_d . We note that, “attractor” has a particular definition in statistical physics, which does not directly translate into the definition outlined in this study.

Let T_{ij} represent the transition probability from cluster i to j at a single day resolution and w_{self} and $w_{attractor}$ represent the parameters associated with self and attractor transitions. The transition probabilities can be summarized as:

$$T_{ij} = \frac{e^{w_{self}(j) \cdot \delta_{ij} + (1 - \delta_{ij}) \cdot (w_{attractor}(j|i))}}{\sum_j e^{w_{self}(j) \cdot \delta_{ij} + (1 - \delta_{ij}) \cdot (w_{attractor}(j|i))}} \quad (\text{Equation 1})$$

where δ_{ij} represents Kronecker delta.

The transition dynamics are simplified to a 2x2 transition matrix per cluster i that represents *self* (i) and *not-self* (i^c) states and the rates of transition between those states:

$$\tilde{T}(i) = \begin{pmatrix} p_{i,i} & p_{i,j^c} \\ p_{j^c,i} & p_{j^c,j^c} \end{pmatrix} \quad (\text{Equation 2})$$

And the self and attractor parameters are formally represented as:

$$\begin{cases} \log\left(\frac{p_{i,i}}{p_{i,j^c}}\right) = w_{self}(i) \\ \log\left(\frac{p_{j^c,i}}{p_{j^c,j^c}}\right) = w_{attractor}(i) \end{cases} \quad (\text{Equation 3})$$

Inference of drug influence on cluster dynamics via elastic net regularized regression

In order to investigate high-resolution cluster dynamics in response to drug exposures, we included samples that were collected 1 day apart between day -14 and 14 relative to HCT from patients in the MSKCC discovery cohort. The criteria for drug inclusion in this analysis are as followed: (1) Drugs were administered via oral/IV routes; (2) Drug exposures occurred between day -14 to 14 relative to HCT; (3) Drugs were administered to at least 5% and no more than 90% patients in the discovery cohort; and (4) Drug exposures occurred in at least 15 patients with single-day resolution samples.

Equations 1, 2, and 3 provide a formal definition of our microbial dynamics in terms of self and attractor transitions. In practice, we took advantage of a logistic regression fit to solve Equation 3 by assuming a binary value for each transition to a given cluster i . We defined the value of 1 for transitions towards cluster i and a value of 0 otherwise.

We used elastic net regularized regression for feature selection to estimate the influence of drug exposures on self and attractor transitions of each cluster, using the R package caret, e1071 and glmnet for model fitting and parameter tuning. Formally, the coefficients that impact transition probabilities were computed as following:

$$\begin{cases} w_{self}(i, t, X) = \beta_{self,i,0} + \beta_{self,i,1}t + \sum_{k \in \text{Drug set}} \beta_{self,i,k} \cdot X_{k,t} \\ w_{attractor}(i, t, X) = \beta_{attr,i,0} + \beta_{attr,i,1}t + \beta_{attr,i,2}d_{ij} + \sum_{k \in \text{Drug set}} \beta_{attr,i,k} \cdot X_{k,t} \end{cases} \quad (\text{Equation 4})$$

The parameters $X_{k,t}$ have a value of 1 in days patients were exposed to drug k and 0 otherwise. The 10-fold cross-validation partitions were pre-specified such that samples from the same patient are always in the same partition using function *trainControl()* and *train()* in R package caret.

Calculation of bacterial response score of drugs

The dynamic model we proposed can estimate the association between a given drug exposure with quantitative microbiome features. The self and attractor parameters predicted in Equation 4 can be used to construct transition probabilities according to

Equation 1. Let the probability of switching from cluster i to j in the absence of any drug exposure be $\pi_{i \rightarrow j}(\text{no drug})$, the bacterial response score of a drug d on taxon y is defined as:

$$C(y|d) = \sum_{ij} (\pi_{i \rightarrow j}(d) - \pi_{i \rightarrow j}(\text{no drug})) \cdot \log\left(\frac{p_j(y)}{p_i(y)}\right) \quad (\text{Equation 5})$$

where $p_i(y)$ represents the average relative abundance of taxon y from all samples in cluster i . A negative score indicates an antibacterial effect of drug d on taxon y . The transition matrix used to estimate antibacterial score was computed at time $t=0$.

The bacterial response scores estimated from Equation 5 were compared to published *in vitro* measurements of anti-bacterial activities.³ The *in vitro* measurements evaluated the effect of specific drugs using the area under the curve (AUC) of bacterial growth. An AUC value significantly below 1 indicates that the drug inhibited bacterial growth *in vitro*. We restricted our analysis to 19 bacteria species that appeared in more than 10% of the MSKCC samples with relative abundance above 10^{-4} . We identified the relative abundance of a given species by summarizing the relative abundance of all ASVs mapped to that species. Our prediction was considered consistent with *in vitro* measurements when bacterial response scores were negative for bacteria-drug pairs that showed inhibition in that study. Statistical significance was evaluated using two-sided Wilcoxon's rank-sum test.

Defining patient-specific bacteria response scores based on drug exposure profiles

A risk score that is associated with patient outcomes can be computed from how drug exposures are predicted to influence the microbial dynamics. We defined specific target values, τ , which are features of the intestinal microbiota associated with each cluster, and adapted Equation 5 to compute a risk score, S , for each patient by averaging the contribution of all drug exposures that occurred within day -14 to 14 relative to HCT.

$$S = \sum_{t \in \text{time}} \sum_{d \in \text{drugs}} \sum_{ij} (\pi_{i \rightarrow j}(d, t) - \pi_{i \rightarrow j}(\text{no drug}, t)) \cdot \log\left(\frac{\tau_j(y)}{\tau_i(y)}\right) \quad (\text{Equation 6})$$

For this work, we considered four target values: relative abundance of *Enterococcus*, *Erysipelatoclostridium*, *Blautia*, and alpha diversity. Patient-specific bacteria response scores for each of the considered microbiome features were calculated based on drug exposure profiles between day -14 and 14 and compared with observed taxa relative abundance or alpha diversity in samples collected between day 14 to 45 relative to HCT using Pearson's correlation, with adjusted p-values by Benjamini-Hochberg's correction. The median values were taken for patients with multiple samples available.

Imputation of missing samples

The dynamic model defined in Equation 1 can be used to simulate microbial dynamics and to impute cluster state in time points without sample collection. Equation 1 defines transition probabilities at single day resolution and forms the fundamental unit to predict microbial dynamics. Let $X(t)$ be the compositional state at time t , the value T_{ij} is equivalent to $P(X(t+1)|X(t))$. Assuming first-order Markov chain, the transition probability matrix between samples at longer time distance can be computed from its fundamental units as following:

$$T(t_0 \rightarrow t_0 + d) = T(t_0) * T(t_0 + 1) * \dots * T(t_0 + d - 1) \quad (\text{Equation 7})$$

Where $T(t_0)$ represents transition probabilities at single day resolution between time t_0 to t_0+1 .

The imputation of cluster states at a given time point can be performed by considering the state at nearby time points. Interpolations can be made based on past samples (forward interpolations), future samples (backward interpolations) or both (forward-backward interpolations). Consider the estimation of a cluster state at time point t from samples available at time point $t-a$ and $t+b$. Forward interpolation imputes the data point based on the probability $P(X_t|X_{t-a})$, backward prediction from $P(X_t|X_{t+b})$ and backward-forward interpolation from $P(X_t|X_{t-a}, X_{t+b})$. Forward interpolations can be estimated from Equation 7. The backward interpolations were obtained by adapting model described in Equation 1 to predict cluster state at time t based on cluster state at time $t+1$ and exposures at time t . The backward-forward interpolation takes advantage of both forward and backward equations and is computed as:

$$P(X_j|X_{j-a}, X_{j+b}) \propto \frac{P(X_j|X_{j+b} = x_b) \cdot P(X_j|X_{j-a} = x_a)}{P(X_j)} \quad (\text{Equation 8})$$

A modified forward-backward interpolation was also tested in which final predictions is obtained by either forward or backward probabilities:

$$P(X_j|X_{j-a}, X_{j+b}) \propto \begin{cases} P(X_j|X_{j+b} = x_b) & \text{if } w(t) > c \\ P(X_j|X_{j-a} = x_a) & \text{if } w(t) \leq c \end{cases} \quad (\text{Equation 9})$$

Where $w(t)$ indicates whether forward or backward interpolation should be used to predict state at time t , and c indicates the mean of the forward-backward prediction probabilities.

The imputation accuracy for each model described above was estimated using 10-fold cross validation, in which 90% of patients were selected as the training set and 10% as the test set. In the training set, the logistic regression model defined in Equation 5 was used to estimate the parameters for *self* and *attractor* weights as well as the decision parameter for modified forward-backward prediction. In the test set, the regression coefficients were used to reconstruct transition matrices and compute forward, backward, forward-backward as well as modified forward-backward probabilities. The microbial state was classified by the cluster with the maximum probability.

Linear mixed-effects model of drug-species associations using shotgun metagenomic profiles

We defined a linear mixed-effects model to investigate the association between drug exposures and changes at a species-level resolution. Species relative abundance was predicted by MetaPhlAn 3.0 using shotgun metagenomic data. The model considers pairs of samples that were collected between day -14 and 14 relative to HCT and less than five days apart, and with a minimum species relative abundance of 10^{-4} . We assumed that changes in species relative abundance depend on time of sample collection and drug exposure. Formally, let a given sample pair be collected at time t_1 and t_2 , respectively. The log difference in species relative abundance, $\ln(Sp_{t2}/Sp_{t1})$ depends on fixed effects of drug exposures and a random effect relative to time of sample collection, t_1 , as following:

$$\ln\left(\frac{Sp_{t1}}{Sp_{t2}}\right)(t_1, t_2, X) = 0 + \beta \cdot X_{t1,t2} + (1|t_1) \quad (\text{Equation 10})$$

The parameter t_1 represents time of first sample collection in the sample pair, binned into weekly intervals relative to HCT, as a random effect variable. $X_{t1,t2}$ represents a given drug exposure and has the value of 1 if patients were exposed to the drug in the time interval t_1 and t_2 . P-values from the linear mixed-effects model were adjusted for multiple hypothesis testing using Benjamini-Hochberg's correction. Equation 10 was solved using R package lme4.

Dominant strain dynamics and their association with drug exposures

StrainPhlAn 3.0 was used to profile bacterial community at the strain-level resolution.^{36,49} This algorithm identifies the most dominant strain per species per sample by reconstructing dominant consensus sequence variants across species-specific marker genes. We applied StrainPhlAn 3.0 to a dataset of 980 shotgun metagenomic shotgun samples in the MSKCC discovery cohort, which returned the multiple sequence alignment of the dominant strain for a given species and the RAxML (Randomized Axelerated Maximum Likelihood) phylogenetic tree across samples. The phylogenetic tree was then used to calculate the dominant strain phylogenetic distance across a pair of samples. Phylogenetic distance is defined as the branch length between two nodes in the StrainPhlAn phylogenetic tree, normalized over the total branch length of the tree. Branch length was calculated using R function *cophenetic()*. We focused on strains from five species of interest, including *Blautia coccooides*, *Blautia producta*, *Enterococcus faecalis*, *Enterococcus faecium* and *Erysipelatoclostridium ramosum*.

For analysis of dominant strain convergence in association with antibiotic and non-antibiotic exposures, we compared the phylogenetic distances of dominant strains within the species *E. faecium* across pairs of samples that were collected less than five days apart between day -14 and 14 relative to HCT. Exposures were considered true if patients received the drug during sample pair collection time. Antibiotic exposure includes the seven investigated antibiotics in this study. Non-antibiotic exposure comprises seven drugs with the highest absolute *Enterococcus*-response score values, indicating strongest associations with changes in *Enterococcus* relative abundance. These non-antibiotic drugs include diphenoxylate/atropine, polyethylene glycol, levothyroxine, fentanyl, methotrexate, anti-thymocyte globulin and cyclosporine. Statistical significance was evaluated using two-sided Wilcoxon's rank-sum test.

Model validation using simulated data

We validated our method using simulated data of known parameters. We considered 62 possible drug exposures and estimated their elastic net regularized regression coefficients. We simulated a dataset of single day resolution samples for 1,000 patients ranging from day -14 to day 14. Drug exposures were simulated with a 50% possibility for each drug at each day. The cluster state at day -14 was randomly sampled according to pre-transplant cluster frequency observed in the MSKCC cohort. Following days were randomly sampled according to transition probability conditioned to known state at preceding day, $P(X_t | X_{t-1}=X_{t-1})$. We used this single-day resolution simulated dataset to feed into the elastic net regularized regression to determine how well the model can recover the pre-defined self and attractor coefficients utilized to simulate cluster states. Performance of our method was evaluated using Pearson's correlation to estimate the correlation between coefficients estimated from real dataset and coefficients recovered from simulated dataset (Figure S7A).

Inference of drug influence on genus relative abundance via elastic net regularized regression

Analysis of changes in specific taxon relative abundance as a function of environmental or clinical factors using regression models is a commonly employed computational approach in microbiome studies. However, this approach focuses on pre-specified taxa, which does not account for the community structure of the intestinal microbiota compositions and has potential mathematical constraints including saturated and overfitted models. By employing a clustering approach to reduce complexity while preserving the microbiome community structure, PARADIGM can potentially overcome these mathematical challenges. To demonstrate the

improved prediction power of PARADIGM, we analyzed changes in each genus relative abundance between daily collected sample pairs as a function of drug exposures and time of sample collection. The dataset and input parameters were identical to the training dataset for PARADIGM (Table S2). Based on this model, we predicted future changes in genus relative abundance between day 14 and 45 and compared with observed values using Pearson's correlation. We demonstrated that while the associations between drug exposures and genus relative abundance were correlated between the single-taxon approach and PARADIGM approach (Figure S7B), PARADIGM could reliably predict future microbiome trajectories (Figure 5C) while the single-taxon approach could not (Figure S7C).

Co-inclusionary and exclusionary relationships between bacteria and cluster stability

To identify the potential microbiome influence on the stability of *Enterococcus* domination in allo-HCT patients, we developed a logistic regression model with lasso penalty. This model takes into account the relative abundance of different bacterial genera on the stability of *Enterococcus*-high cluster 10. We defined cluster 10 stability as the rate at which the microbiota compositions stay in the same cluster 10, assuming a value of 1 when patients stay in cluster 10, and a value of 0 when patients move from cluster 10 to any other cluster among a pair of daily collected samples. We included as parameters antibiotics exposure, day of sample collection relative to HCT, alpha diversity and relative abundance of top 20 most abundant genera in cluster 10. Antibiotic exposure includes exposure to any of the seven antibiotics investigated in this study. The input dataset consists of pairs of daily samples collected between day -14 and 100 relative to HCT. Formally, the coefficients of these potential drivers of cluster 10 stability were computed as following:

$$w_{\text{self}}(i, t, \text{abx}, p) = \beta_{\text{self},0} + \beta_{\text{self},1}t + \beta_{\text{self},2}\text{abx}_t + \beta_{\text{self},3}\text{simpson_reciprocal}_t + \sum_{i \in \text{Top 20 genes}} \beta_{\text{self},j} * p_{i,t} \quad (\text{Equation 11})$$

In which $p_{i,t}$ represents the relative abundance of each taxon in the top 20 most abundant genera in cluster 10, and abx indicates if patients received any of the seven investigated antibiotics in this study on day t . Coefficient values indicate the direction and magnitude of association between each parameter and the stability of *Enterococcus*-high cluster 10, with a negative value indicating a negative association with *Enterococcus* domination stability.

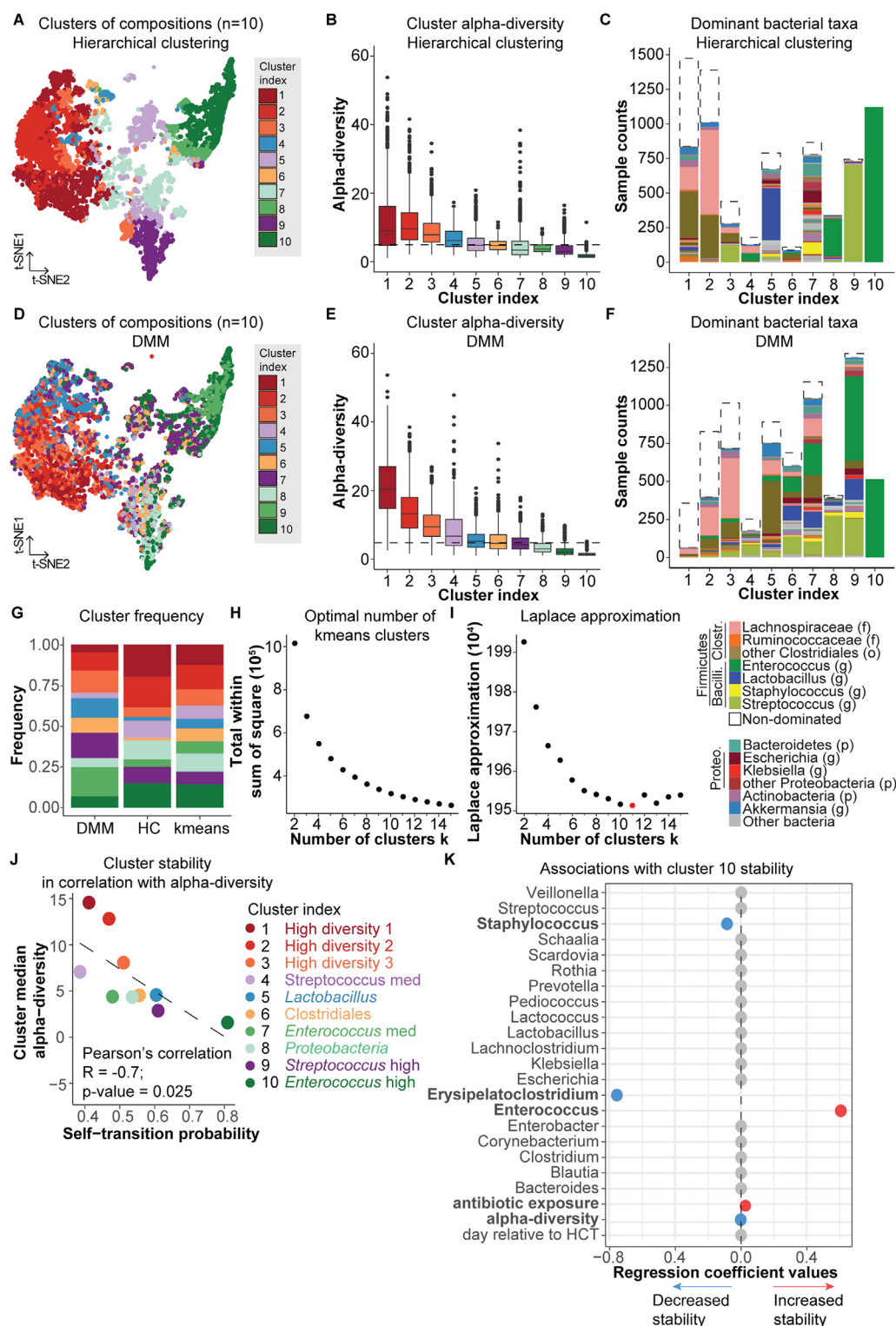
Statistical analyses of survival outcomes

We used landmark analyses of survival beyond day 14 relative to HCT using R package survival. Patients were censored at the time of last contact or at the time of second allo-HCT (when applicable). All survivors were censored at two years of follow-up. Patients randomized to the FMT arm of the trial were excluded from analysis of clinical outcomes. Patient-specific bacteria response scores corresponding to a given microbiome feature were considered as a continuous variable in a multivariate Cox proportional hazard model, controlled for age, sex, conditioning intensity, graft source and underlying disease as variables with R function `coxph()`. P-values were adjusted for multiple hypothesis testing using Benjamini-Hochberg's correction.

Competing risk analyses were performed to identify the association between patient-specific bacteria response scores and cause-specific mortality. We investigated three competing events, namely relapse (defined here as relapse or progression of disease), GVHD-related mortality (defined here as death due to GVHD or after GVHD onset, without relapse), and transplant-related mortality (encompassing deaths from GVHD, infections and organ toxicities). For each competing event, multivariate Fine-Gray subdistribution hazard models were fit by R function `crr()` from R package `tidymprks`. Hazard ratios are presented in main text with the 95% confidence interval indicated in parentheses.

To tease out the potential causal relationships between medication, microbiome and mortality, we compared the effect sizes and statistical strengths of patient-specific bacteria response scores and observed genus relative abundance or diversity values in predicting all-cause mortality in the MSKCC and Duke validation cohorts. In order to assure proper magnitude comparison, microbiome measurements were rescaled by Z-score normalization and fit into either an independent (which consider either microbiome metrics or response scores) or a competing (which considers microbiome feature and response score together) multivariate Cox proportional hazard model, controlled for age, sex, conditioning intensity, graft source and underlying disease. The independent and competing models consider only patients with stool samples collected between day 0 and 45 relative to HCT (MSKCC validation cohort: 340 patients; Duke validation cohort: 108 patients). Genus abundance was calculated from stool samples collected between day 0 and 45 relative to HCT, response scores were calculated using drug exposure profiles between day -14 to 14 relative to HCT. The median values were taken for patients with multiple samples available. P-values were adjusted for multiple hypothesis testing by Benjamini-Hochberg's correction.

Supplemental figures



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Figure S1. Clusters of intestinal microbiota during allo-HCT were identified by other unsupervised clustering methods, related to Figure 1

(A and D) Compositional space of the intestinal microbiota visualized by tSNE projection in the MSKCC discovery cohort. Each dot represents a sample, colored according to clusters assigned by (A) hierarchical clustering or (D) Dirichlet multinomial mixture (DMM) model.

(B and E) Alpha diversity per cluster identified by (B) hierarchical clustering or (E) DMM. The horizontal dashed line represents the median alpha diversity of samples in the MSKCC discovery cohort. Box plot includes a mid-line, upper hinge, and lower hinge, which represent the median and third and first quartiles, respectively.

(C and F) Compositional characteristics per cluster identified by (C) hierarchical clustering or (F) DMM. Samples in which the relative abundance of the most common taxon is at least 30% are color-coded by the most common taxon. Non-dominated samples are colored white. p, phylum; f, family; o, order; g, genus.

(G) Distribution of samples across 10 clusters identified by three unsupervised clustering methods.

(H) Optimal number of k-means clusters was estimated from the curve of within-cluster sum of square distances from each point to its cluster centroid.

(I) Optimal number of clusters identified by DMM was estimated by the smallest Laplace approximation metric.

(J) Correlation between cluster stability and alpha diversity. Cluster stability was measured by self-transition probability. Alpha diversity is defined as the cluster median reciprocal Simpson diversity.

(K) Co-exclusionary and inclusionary relationships associated with the stability of *Enterococcus*-high cluster 10.

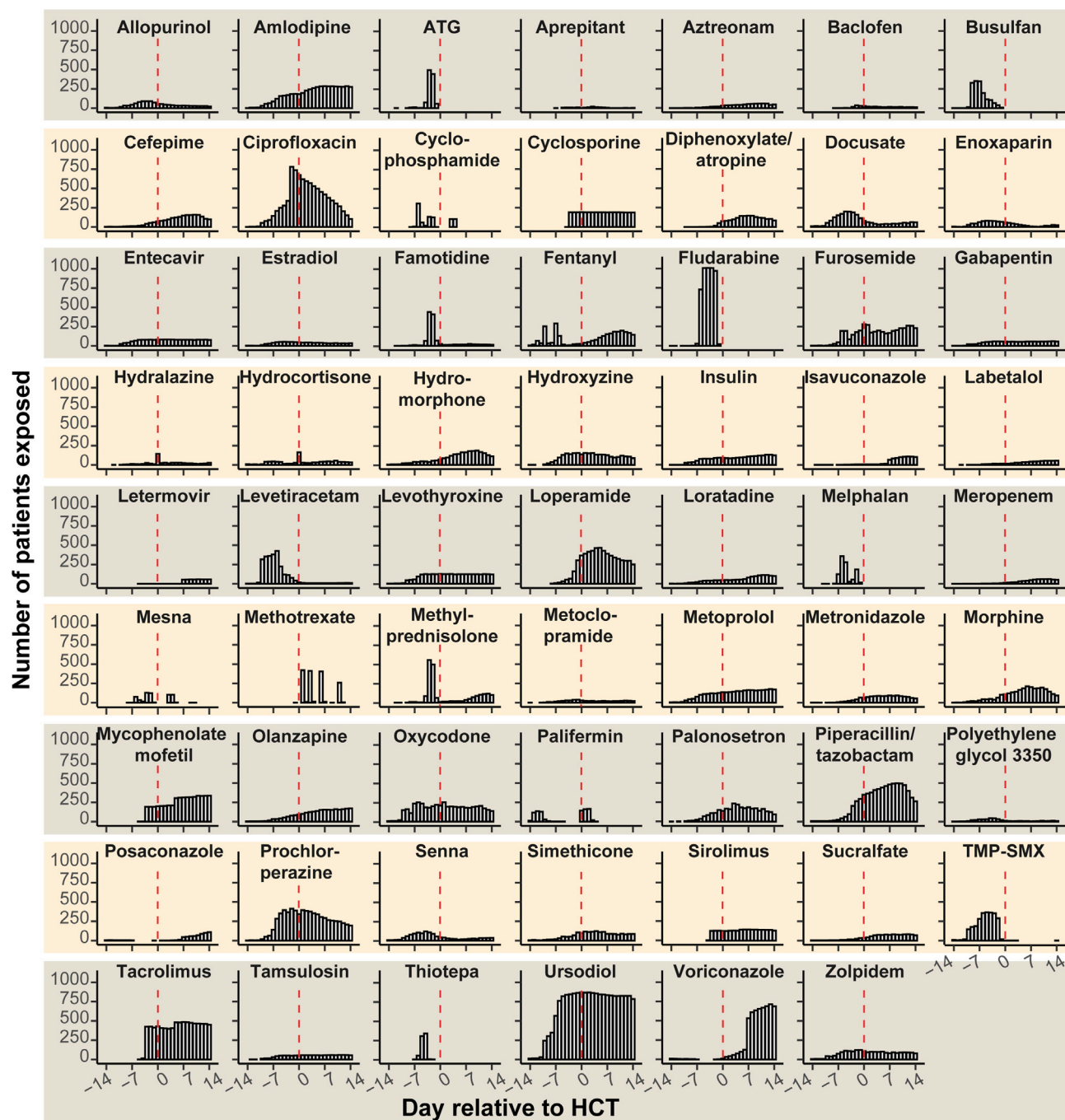
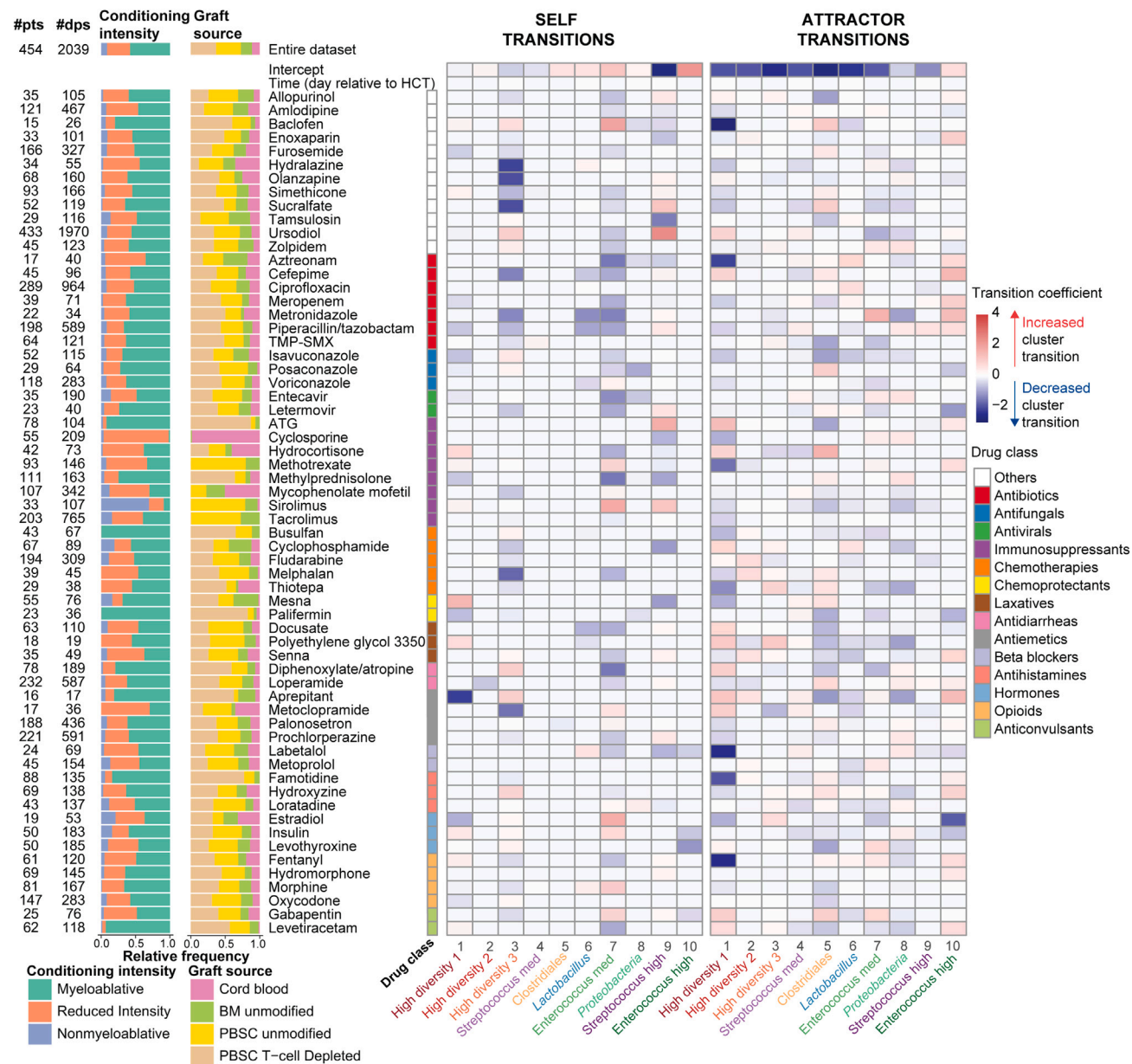
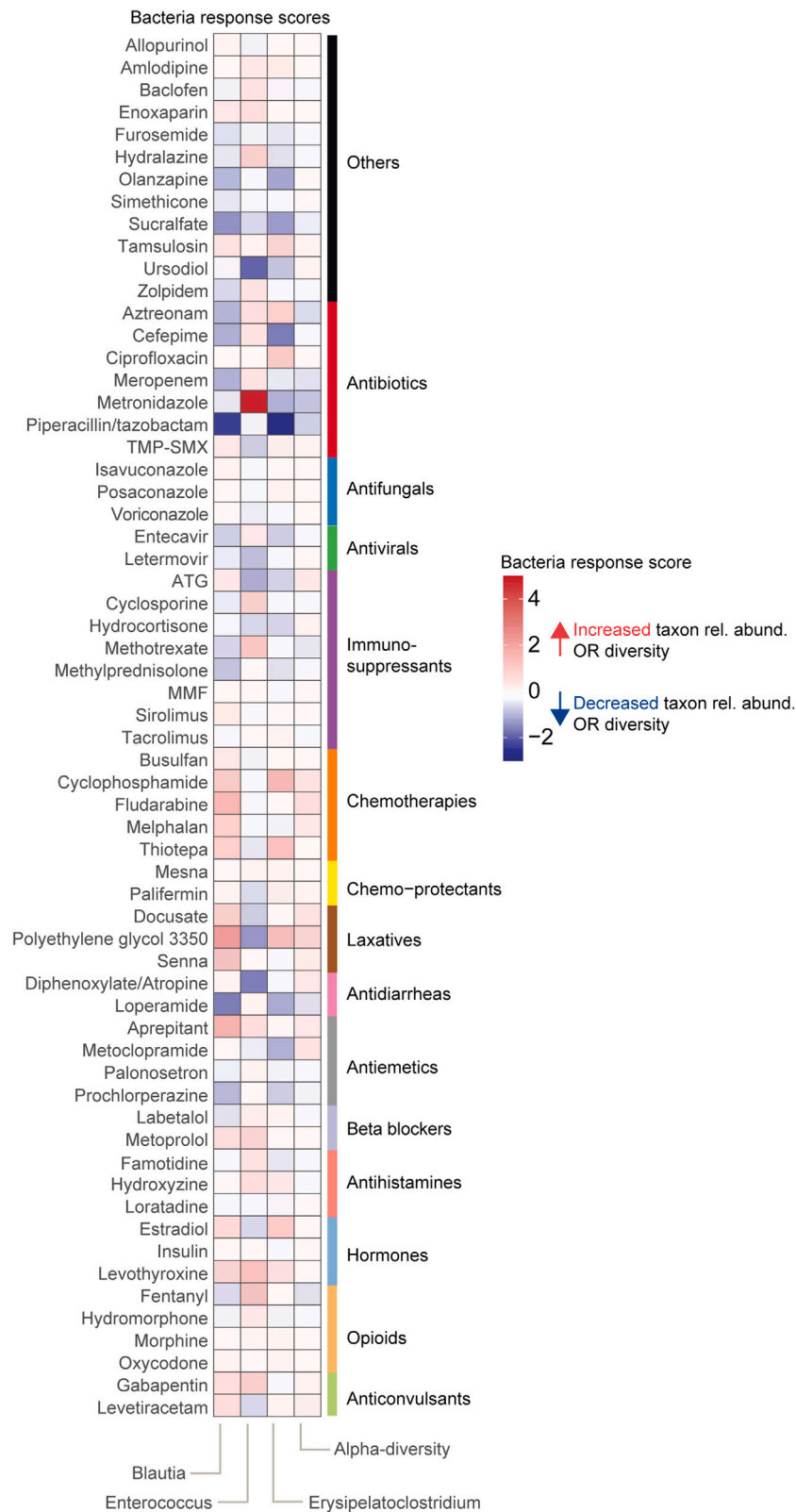


Figure S2. Time courses of drug exposures between day -14 to 14 relative to allo-HCT, related to Figure 3
Red dashed lines indicate day 0, which is the day of stem cell infusion.





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Figure S4. Bacteria response scores for four microbiome features of interest, related to Figure 3

Bacteria response scores predict the association between a given drug exposure and changes in genus relative abundance or alpha diversity. Positive response scores (red shades) indicate that drug exposures are associated with increased genus relative abundance or alpha diversity. Negative response scores (blue shades) indicate that drug exposures are associated with decreased genus relative abundance or alpha diversity.

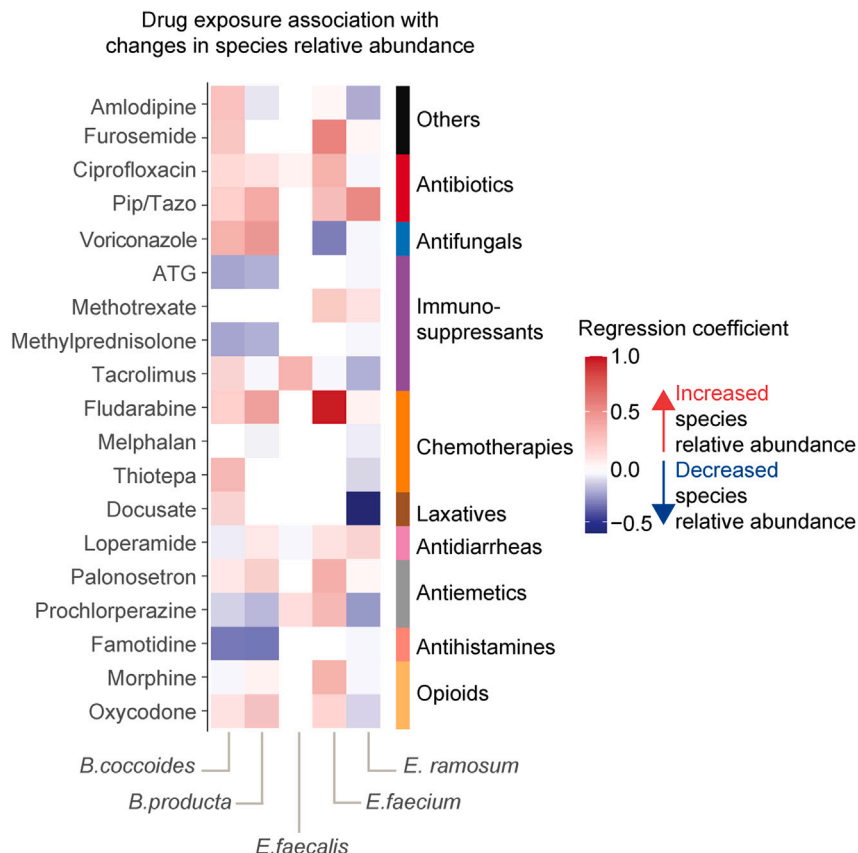


Figure S5. Associations between drug exposures and changes in species relative abundance from samples with shotgun metagenomic sequencing, related to Figure 4

Log changes in species relative abundance between subsequently collected samples were analyzed as a function of individual drug exposure in a linear mixed-effects regression model, with time of sample collection binned into weekly intervals as a random effect variable. Positive coefficient values (red shades) indicate that drug exposures are associated with increased species relative abundance. Negative coefficient values (blue shades) indicate that drug exposures are associated with decreased species relative abundance. A white box indicates that there are insufficient data points to fit the regression model for a given drug-species pair.

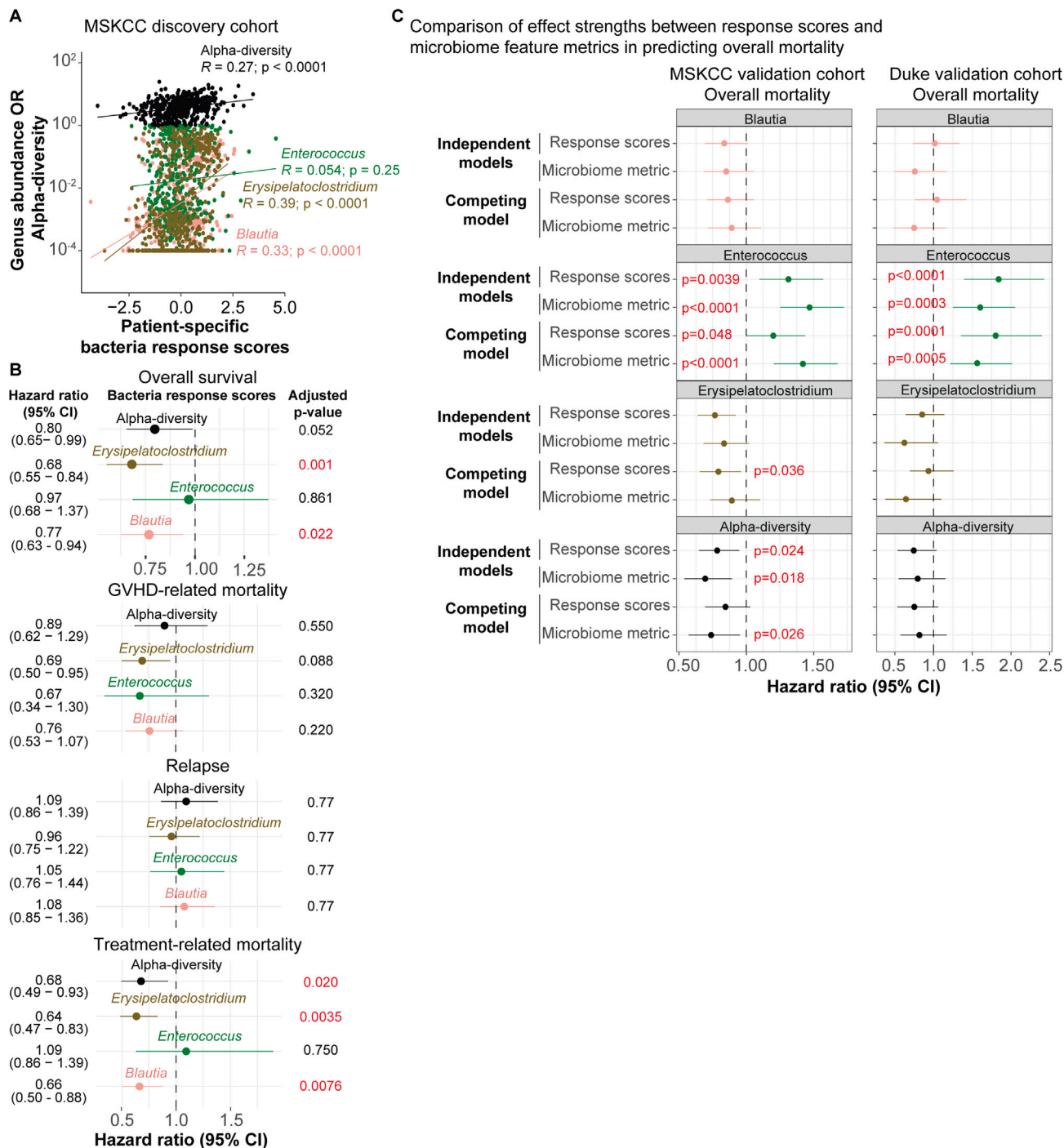


Figure S6. Investigation of causal relationship between drug exposures, microbiome, and mortality in all patient cohorts, related to Figure 5

(A) Pearson's correlation between patient-specific bacteria response scores and observed genus relative abundance or alpha diversity in samples collected between days 14 and 45 relative to HCT in the MSKCC discovery cohort (454 patients, including recipients of PBSC T-cell depleted graft).

(B) Prediction of overall and cause-specific mortality in the MSKCC discovery cohort (295 patients, excluding recipients of PBSC T-cell depleted graft) based on patient-specific bacteria response scores for four microbiome features of interest, in each respective multivariate Cox proportional hazard or Fine-Gray model, controlled for age, sex, conditioning intensity, graft source, and underlying disease. Adjusted p values by Benjamini-Hochberg's correction.

(C) Microbiome feature metrics (taxa relative abundance or alpha diversity) and corresponding response scores were compared in terms of their associations with overall mortality risk. Measurements were fit into either an independent (which considered either microbiome metrics or response scores) or a competing (which considered microbiome metrics and response scores together) multivariate Cox proportional hazard model.

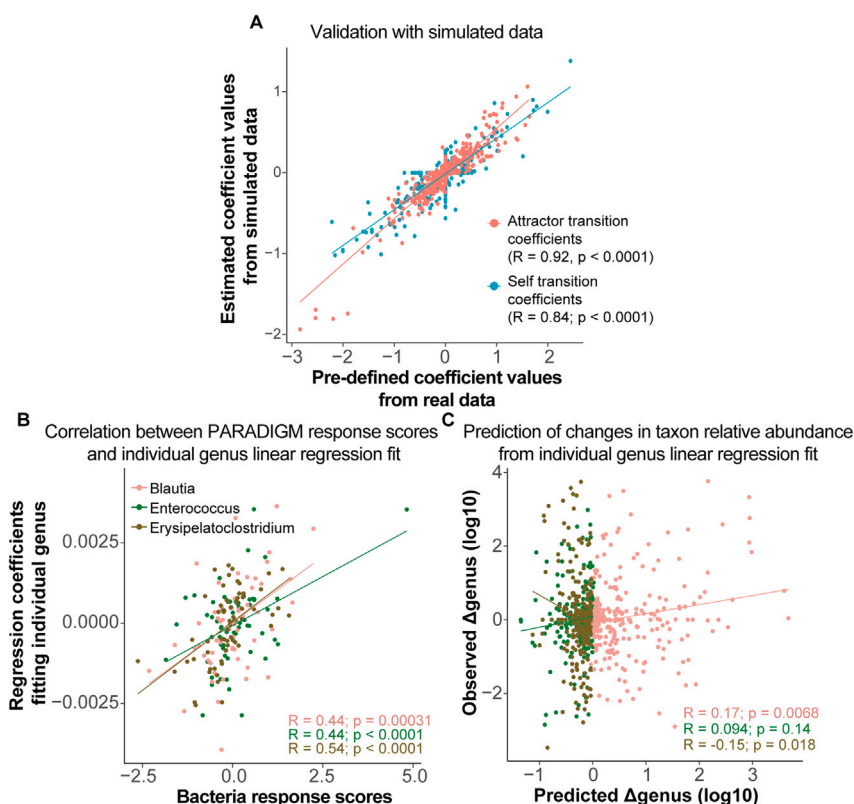


Figure S7. PARADIGM was validated against simulated data and different regression models, related to STAR Methods

(A) PARADIGM was validated against simulated data using Pearson's correlation. This single-day resolution simulated dataset was input into the elastic net regularized regression to determine how well the model can recover the pre-defined self and attractor coefficients utilized to simulate cluster states.

(B) PARADIGM was validated against a separate elastic net regularized regression using drug exposures and time of sample collection as predictors of changes in a single genus relative abundance, using the same training dataset.

(C) Single taxon-based regularized linear regression approach poorly predicted observed changes in *Enterococcus* relative abundance in samples collected between days 14 and 45.